

Project Management (MGMT3192)

YIRGA G...



CHAPTER ONE

General Introduction





Learning project management can help you complete projects **on time, on budget, and on target.**

Project management is **not just** for **project managers**. **Team members** need to know how carry out their parts of the project and **executives** need to understand **how to support** project management efforts in the organization.

Projects help us make desired changes in an **organized manner** and with reduced **probability of failure.**





Faster, cheaper, and better have become the concern of not only profit-making organizations seeking to increase market share and profits but also non profits and governmental organizations seeking to **increase their value to clients.**

Companies are **always in transition** now, remodeling and **reorganizing to meet the latest global challenges.**

Competition is keen and only the **flexible** ones will **survive.**

These business conditions translate directly to the **greater demands** for efficient and effective management of an entire spectrum of **projects.**

Organizations are **increasingly using projects to meet their goals.**



What Is a Project?

A project is a **temporary** endeavor undertaken to create a **unique** product or service.

Temporary – Definitive beginning and end.

A project has a definite **beginning** and definite **end**.

Each activity, and the entire project, has a start and stop date.

The duration of a project is **finite**.

Unique – New undertaking, unfamiliar ground.

A project **has never happened** exactly in the same way **before**, and it **will never happen again** under the same conditions.

Projects involve doing something that **has not been done before** in the same environment.



A project is an **investment** activity in which financial resources are expected to create **capital assets** that produce **benefits** over an extended period of time.

Projects can be large or small and take a short or long time to complete.

A project **ends** when its objectives have been **reached**, or the project has been **terminated**.





Examples of Project

- **Developing a new product or service.**
 - **Constructing a building.**
 - **Construction of a bridge or a road.**
 - **Effecting change in a structure of an organisation.**
 - **Designing a new vehicle.**
 - **Developing or acquiring a new or modified information system.**
 - **Building a water system for a community.**
 - **Organizing a meeting (Hosing conference).**
 - **Planning a wedding.**
- 



A major misconception about a project is relating it to only **construction or creation of such **physical facilities** as buildings, roads and dams. However, project may involve **intangible** things such as**

- Creation of awareness
 - Eradication of diseases
 - Combating harmful practices
 - Capacity building
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Project Features

□ Temporary

A project has a definite beginning and definite end.

Each activity, and the entire project, has a start and stop date.

The duration of a project is finite.

□ Unique

A project has **never happened exactly in the same way before, and it will never happen again under the same conditions. Exact project has not been done before**

Projects involve doing something that **has not been done before in the same environment.**



❑ Progressive elaboration

Project is developed using progressive elaboration (A project occurs step by step to define the product or service).

❑ Requires resources

❑ Have a primary customer or sponsor

The project sponsor usually provides the direction and funding for the project.

❑ Uncertainty

Involves uncertainty (Detail requirements may not be known in advance).

❑ Integration

Project Requires resources to be brought together in a process.



Project Vs Program

Programs and projects serve as instruments for the execution of **development plans** in order to boost the national economy.

A program is a collection of related projects.

Programs comprise multiple projects, they are larger in scope than a single project.

For instance, a health program may include a water project as well as construction of a health center; both aimed at improving the health of a given community that previously lacked easy access to these essential facilities.



Note that projects can stand alone without being part of certain program.

Projects, which are not linked with others to form a program, are sometimes referred to as “stand alone” projects.



The hierarchical relationship among development plans, programs, projects, tasks, and work packages is depicted below:

Development Plans



Programs



Projects



Tasks



Work packages

What is Project Management?

Project management is the application of knowledge, skills, tools and techniques to project activities to meet project requirements.

Project management is the art of planning, organising, leading, reporting and completing a project through people.

All management functions are required to effectively and efficiently manage resources required to complete the project.

The objective of project management is to optimize project cost, time, and quality.

How Does Project Management Benefit you?

- ☞ You will have **goal clarity** and **measurement**.
- ☞ Your **resources** will be coordinated.
- ☞ Your **risks** will be identified and managed.
- ☞ You will increase the possibilities of **time savings**.
- ☞ You will increase the possibilities of **cost savings**.
- ☞ You will increase the possibilities of **achieving the agreed outcome**.
- ☞ You will increase the possibilities to **deliver projects successfully**.

Project Manager

The project manager is the **person assigned to manage a specific project**, who is expected to meet the approved objectives of a project, including project scope, budget, and schedule.

The project manager has **overall responsibility** for planning, organizing, integrating, controlling, leading, decision-making, communicating, and building a supportive climate for the project.

It is the responsibility of the project manager to make sure that the ***customer is satisfied*** that the work scope is completed in a **quality manner, within budget, and on time**.

Team members. The people who work with the project manager to accomplish project goals and complete project activities.

A Good Project Manager

- **Takes ownership of the whole project**
- **Is proactive not reactive**
- **Adequately plans the project**
- **Is Authoritative**(able to be trusted as being accurate or true; reliable.) (**NOT Authoritarian**)
- **Is Decisive**(having or showing the ability to make decisions quickly and effectively)
- **Is a Good Communicator**
- **Manages by data and facts**
- **Leads by example**
- **Has sound Judgement**
- **Is a Motivator**
- **Is Diplomatic**
- **Can Delegate**

Critical Success Factors of A project

Factors that are critical to project success:

- 1. Clearly defined goals and project mission**
- 2. Top management support**
- 3. A competent project manager**
- 4. A competent project team**
- 5. Sufficient resources**
- 6. Customer involvements and consultation**
- 7. Good communication**
- 8. Responsiveness to customers**
- 9. Proper monitoring and feedback**
- 10. Appropriate technology**



CHAPTER TWO

Project Life Cycle



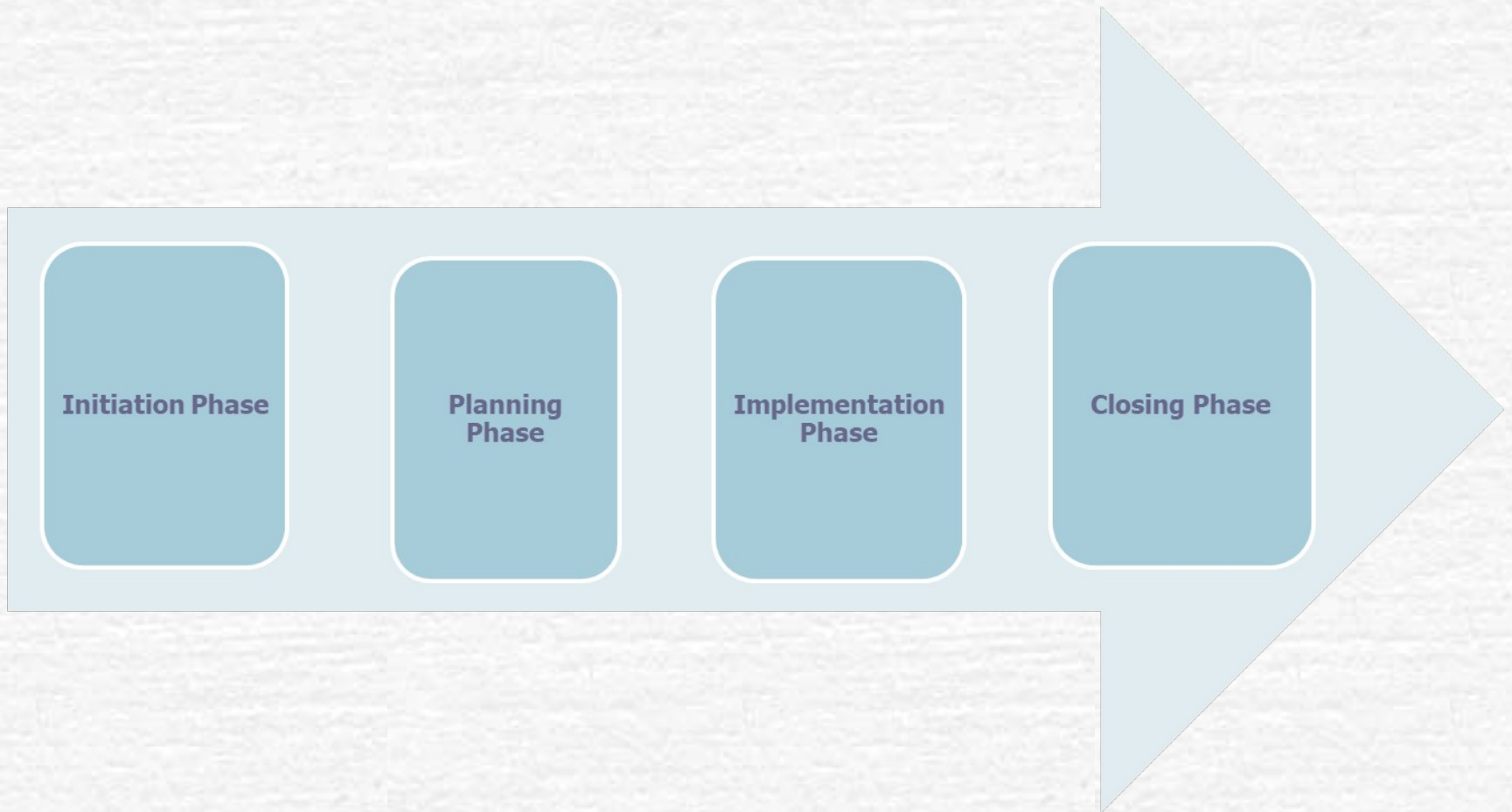
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- ❖ Before any project is actually realized it goes through **various phases**.
 - ❖ Therefore, the different phase through which a project passes constitutes what is often called **“the project cycle”**.
 - ❖ Project cycle is **logical sequence of activities to accomplish the project’s goals and objectives**.
 - ❖ A project cycle is **a sequence of events, which a project follows**.
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Phases of project life cycle

1. **Initiation phase**
2. **Planning phase**
3. **Implementation phase**
4. **Closing phase**

❖ A clear understanding of these phases allows **managers and executives to maintain control of the project more efficiently.**

Project Life Cycle



Initiation Phase

- ❖ This can also be referred to as the **Conceptualization Phase** and is the **starting point of any project or idea.**
- ❖ The phase where a **problem** that the project is trying to solve is **identified** and **potential solutions are suggested.**
- ❖ **Define the Problem** (What is the problem?): Identify the **problem to be solved** by the project. What client **need** is being **satisfied** by the project? Will the development of a project solve that problem? It helps to visualize the desired end result.



Develop Solution Options: How many **different ways** might you go about solving the problem? Of the available alternatives, which do you think will **best solve** the problem?

This is also the phase where the project's **value** and **feasibility** are measured.



Planning Phase

- ❖ Once the project receives the **green light**, it needs a **solid plan to** guide the team, as well as keep them on time and on budget. We should recognise that adequate project planning is **essential**.
- ❖ It is the phase where a project **plan is develop**.
- ❖ In this phase, the project team identifies all of the **work to be done**, prepared the **schedule**, and estimated the **costs**.
- ❖ **Adequate planning leads to the correct completion of work. Inadequate planning leads to frustration towards the end of the project and poor project performance.**

Implementation Phase(Execution)

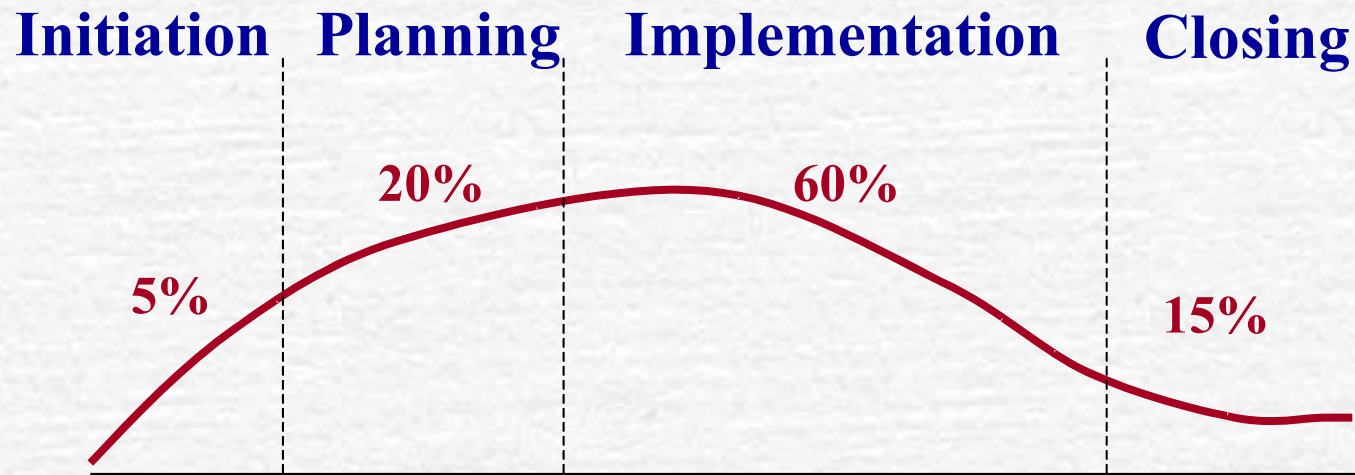
- ❖ This can also be referred to as the **Execution Phase**.
- ❖ Once the plan is developed, it must be **implemented**.
- You can achieve your end result successfully through implementing developed plans.
- During project implementation, **people are carrying out the tasks, and progress information is being reported regularly.**
- Progress is continuously **monitored** and appropriate **adjustments** are made .

If a **deviation** from the plan is discovered, you must ask what must be done to get back on track, or—if that seems impossible—how should the plan be modified to reflect new realities?

Closing Phase

- ❖ The final phase is also called **Termination Phase**.
- ❖ Conclude a project **efficiently and effectively**.
- ❖ During this phase, the emphasis is on passing on the project to customer, handing over project documentation, terminating contracts, communicating the closure of the project to all stakeholders and Preparing the lessons learned document.

Project Life Cycle



Percentages and graph refer to the amount of *effort expended*.

Some models of project cycle

1. world Bank
2. DEPSA's Project Cycle
3. UNIDO

The world Bank (Baum) Project Cycle

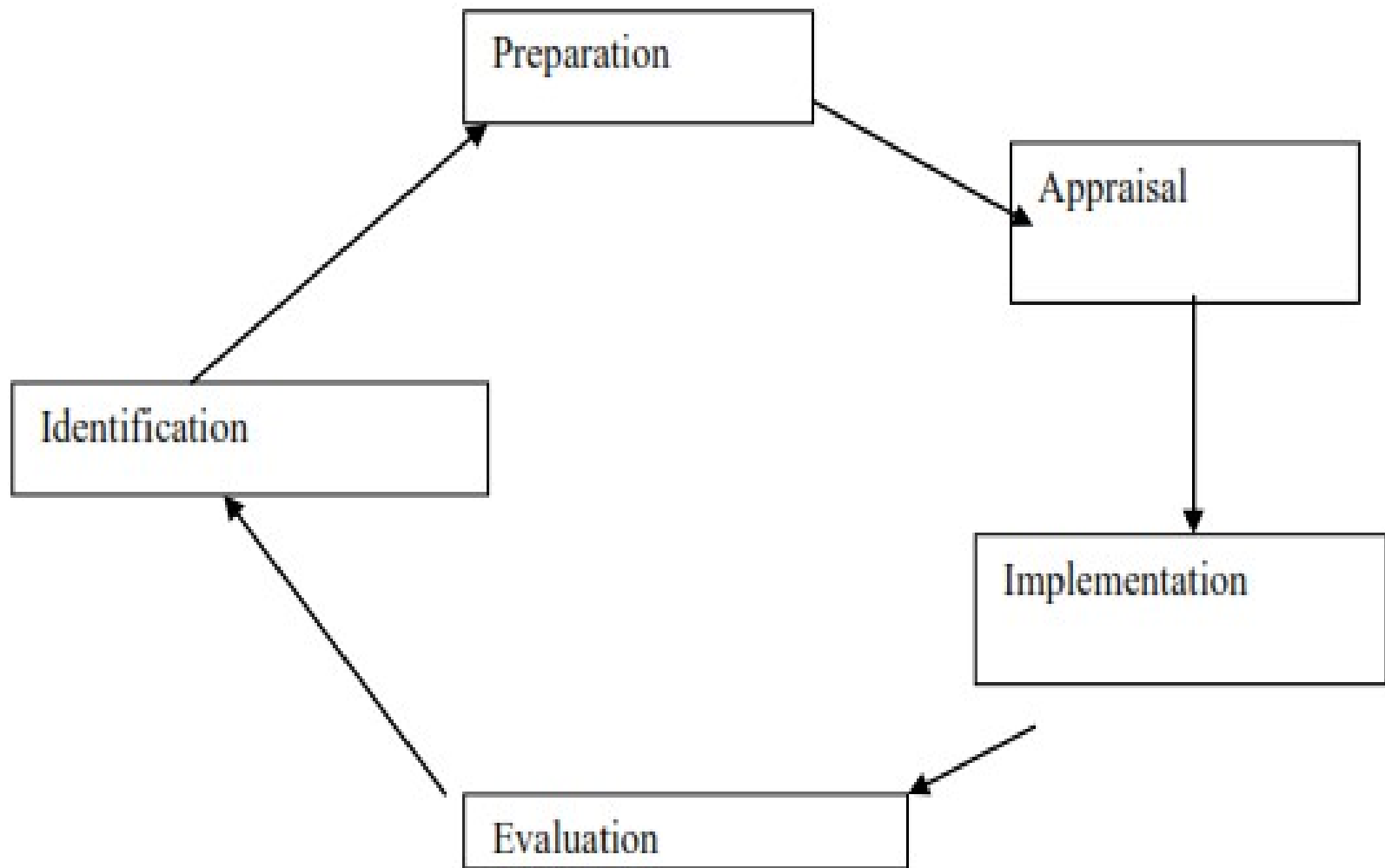
- ❖ The first basic model of a project cycle is that of **Baum model** (1970), which has been adopted by the World Bank
- ❖ The World Bank lends to different countries to Projects ranging from **infrastructure, education, health and government financial management**.
- ❖ In practice, the World Bank and the borrowing country **work closely** throughout the project cycle although they have different roles and responsibilities.



The world bank suggested the following stages:

1. Identification
 2. Preparation
 3. Appraisal
 4. Negotiation and approval
 5. Implementation
 6. Evaluation
- 

The Baum cycle (adapted by the world bank in 1970)



1. Identification

The borrowing country and Bank analyze development strategies **generate project ideas**.

Projects are identified that are financially, economically, socially, and environmentally sound.

Based on sector work and country strategies, the World Bank and borrowing countries **jointly identify projects** that support their development goals.



The first stage in the cycle is to find potential projects.

Some sources of projects are given here.

- Some may be “**resource based**” and stem from the opportunity to make profitable use of available resources.
 - Some projects may be “**market based**” arising from an identified demand in home or overseas markets.
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Others may be “**need-based**” where the purpose is to try to make available to all people in an area of minimal amounts of certain basic material requirements and services.

Well informed technical specialists and local leaders are also common sources of projects.

- Technical specialists will have identified many areas where **they feel new investment** might be profitable,
- local leaders may also have suggestion about where **investment might be carried out.**



✓ Ideas for new projects also come from **proposals** to extend existing programs.

In general, most projects start as an elementary idea.

Eventually, some simple ideas are elaborated into a form to which the title “project” can be formally applied



2. Preparation(pre – feasibility or feasibility studies)

- ❖ The borrowing country is responsible for project preparation.
- ❖ Clients conduct studies and impact assessments that refine the objectives, components, schedule, institutional responsibility and prepare final project documentation.
- ❖ The Bank, on the other hand, starts to determine the conditions required for the project to succeed and for the Bank to be satisfied the project will have positive economic, financial, social and environmental impacts.



Once projects have been identified, there begins a process of progressively more detailed preparation and analysis of project plans.

At this stage the project is being **seriously considered** as a definite investment action.

Project preparation (project formulation) covers the establishment of;

- **A. Technical**
 - **B. Economic and**
 - **C. Financial feasibility**
- 



Project design and formulation is an area in which local and international consultants are very active especially for big project that cover large areas and have big budgets

3. Appraisal

- The Bank is solely responsible for project appraisal.
- Bank staff review the work done during identification and preparation.
- They will review all the studies conducted in previous stages. such as

- 
- After a project has been prepared, it is generally appropriate for a *critical review* or an independent appraisal to be conducted.
 - This provides an opportunity to **re-examine** every aspect of the project plan to assess whether the proposal is appropriate and sound before large sums are committed.
- 



Appraisals should cover at least **seven aspects** of a project, each of which must have been given special consideration during the project preparation phase:

- **Technical** – here the appraisals concentrate in verifying whether what is proposed will work in the way suggested or not.
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- **Financial** – the appraisals try to see if the requirements for money needed by the project have been calculated properly, their sources are all identified, and reasonable plans for their repayment are made where necessary.
- **Commercial** – the way the necessary inputs for the project are conceived to be supplied is examined and the arrangements for the disposal of the products are verified.

✓ **Incentive** – the appraisals see to it whether things are arranged in such a way that all those whose **participation** is required will find it in their interest to take part in the project, at least to the extent predicted in the plan.

✓ **Economic** – the appraisal here tries to see whether what is proposed is good from the viewpoint of the national economic development interest when all project effects (positive and negative) are taken into account and check if all are correctly valued.

- **Managerial** – this aspect of the appraisal examines if the capacity exists for operating the project and see if those responsible ones can operate it satisfactorily. Moreover, it tries to see if the responsible are given sufficient power and scope to do what is required.
- **Organizational** – the appraisal examines the project if it is organized internally and externally into units, contract policy institution, etc so as to allow the proposals to be carried out properly and to allow for change as the project develops.

4. Negotiations and Board approval

After project appraisal, negotiations between the Bank and the borrower country enter the final phase.

During this phase, the bank and borrower come to an **agreement** on the terms of the loan and the project is **presented to the Board for approval**.

5. Implementation

Project implementation is the **responsibility** of the **borrowing country**, while the Bank is responsible for **supervision**.

The World Bank's role is to **monitor** project implementation to ensure that the terms of the loan/credit agreement are followed and that procurement is conducted according to the World Bank's guidelines.



The objective of any effort in project planning and analysis clearly is to have a project that can be **implemented to the benefit of the society**.

Thus, implementation is perhaps the most **important part** of the project cycle.

Recording, monitoring and progress reporting are important activities during the implementation stage.

project implementation must be **flexible**



6. Evaluation

Following completion of the project, the Bank's **Independent Evaluation Group** conducts an **audit** of the project, where the project's outcome is measured against its original objectives.

Reports are then submitted to the **executive directors** and **the borrower**.

2. DEPSA's Project Cycle

- According to the Guidelines to project planning in Ethiopia (1990) of **Development Project Studies Authority (DEPSA)**, the project cycle comprises three major phases.
 - Pre – investment
 - Investment and
 - Operation

Each of these three phases may be divided into stages. The Guideline has divided the Project cycle into six stages

- Identification
- Preparation
- Appraisal/decision
- Implementation
- Operation
- Ex-post evaluation

The pre – investment phase consists of the **first–three stages**, the investment phase includes the **fourth** stage and the operation phase covers the last two stages

The UNIDO Project Life Cycle

The United Nations Industrial Development organization (UNIDO) is the specialized agency of the United Nations that promotes industrial development for poverty reduction, inclusive globalization and environmental sustainability.

The following phases are the project life cycles according to UNIDO:

1. Pre-investment phase
2. Investment Phase
3. Operation phase



1. Pre- Investment Phase

- ☛ Opportunity study(identification of project ideas)
 - ☛ Pre-feasibility study (preliminary project formulation, selection of alternatives)
 - ☛ Feasibility study (techno-economical project back ground, final project formulation stage)
 - ☛ Evaluation report (decision making about project availability)
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2. Investment Phase

Investment (implementation) phase involves

- ✓ Technological acquisition and transfer
- ✓ Detailed engineering design and contract, including tendering, evaluation of bids and negotiations
- ✓ Acquisition of land, construction work and installation
- ✓ Pre-production marketing, including the securing of suppliers and setting up the administration of the firm
- ✓ Recruitment and training of personnel and
- ✓ Start-up

3. Operation phase

- ✓ Replacement of equipment
- ✓ Liquidation



CHAPTER THREE

Project Identification





Project identification is the selection of projects, which are acceptable for financing.

Every project starts with an idea, which is provoked by a need or a problem that has been insufficiently solved or not solved at all.

It begins with the conceiving of ideas. These ideas are then transformed into a project.

Most project ideas will abandon at a very early stage.



Project ideas conceived by:

- Individuals
- Groups of individuals (community)
- Local leaders
- NGOs
- Policy makers
- Planners
- Government organizations
- International development agencies



Sources of Project Ideas

It is important to **have a great idea** first. Project ideas are generated through different sources.

Organizations should try to enhance **people's creativity**, scan the entire business environment and appraise the company's **strengths and weaknesses** to generate a large number of ideas.





In order to discuss the sources of project ideas it is important to know the **industry in which the project** is taking place.

For example, the source of project ideas in a school could come from the **students and/or teachers**. In the workplace, project ideas could come from **managers, team leaders** or **other employees**. Other sources come from **yourself** (an individual), reading, interest in a topic or problem, and talking with other people.





Organizations should also analyze the business environment that consists of the **economic** sector, the **governmental** sector, the **technological** sector, the **socio-demographic** sector, the **competition** sector and the **supplier** sector.





Sources of Project Ideas

- Development plans made by government
 - Sectoral strategies and programmes
 - Surveys conducted by local government and regional organizations
 - Reviews of past projects
 - Private sector plans
 - R & D (Research and Development) department
 - Multilateral or bilateral development agencies
 - Unusual events such as droughts, floods, earth – quakes, hostilities, etc.
 - Workshops
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- Universities
 - Development experiences of other countries
 - The desire of local community to enhance their economic status and improve their welfare
 - Unsatisfied demand or needs
 - The existence of unused or underutilized resources
 - Advertisings
- 



In addition,

- Individual inspiration
 - Employees
 - Customers
 - competitors
 - Distributors, Suppliers, may generate some interesting project ideas.
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Screening Project Ideas

- ☛ Once a list of project ideas has been generated (put forward), they should be carefully screened.
- ☛ Select one or more **potentially promising** project idea(s).
- ☛ This, calls for a quick preliminary screening by **experienced professionals** who could also modify some of the proposals.



During the screening to **eliminate ideas**, which are **not promising**, it is required to look into the following aspects:

- Coherence with strategic priorities of **implementing agencies** and **donors**
 - Consistency with **government priorities**
 - Availability of **inputs**
 - Adequacy of **market**
 - Reasonableness of **costs**
 - Acceptability of **risk level**
- 



During selection, the analyst should eliminate project ideas that:

- Are **risky**,
 - Have **no market** for the output,
 - Have **inadequate supply** of inputs,
 - Are **very costly** in relation to benefits,
 - Assume **over ambitious** sales and profitability.
- 



Methods for Identifying New Project Ideas

Interview: Interviews with a fixed set of (open) questions.

Focus group discussion: Interviews can also be conducted with groups. Arguments expressed in groups offer valuable insights and allow a great deal of information to be collected in a short time.

Workshop:

Brainstorming and SWOT Analysis are two techniques widely used in workshops.





Brainstorming is a creativity tool, which was developed to elicit new, fresh ideas and as such it is very suitable for looking for opportunities and solutions to problems. It works best in medium-sized groups of 8 to 20 people that are familiar with the situation and have a genuine interest.





CHAPTER FOUR

Project Preparation





Project preparation is a study to **decide** whether the identified project is **attractive enough to go** for implementation.

Project Preparation consists of all the work necessary to ensure that a proposed project is **feasible** and **appropriate** and that it can be **successfully implemented**.

This stage involves a more **thorough exercise** of **collecting data and information** on the proposed project.

The exercise is performed by personnel **with technical skills** in consultation with the beneficiary community.



Why is Preparation Important?

Systematic and effective Project Preparation is important for a range of reasons outlined below:

- Project **risks** are managed and controlled.
- Scarce implementation **resources** are **optimally utilized** and are only allocated to viable projects.
- Provides **valuable information** for a “go/no-go” decision
- Narrows the business alternatives.
- Identifies a **valid reason** to undertake the project.
- Projects are supported by the **key stakeholders** (including the community, municipality, funders and implementation partners).
- Government and other funders can **predict** and therefore **manage their cash flows** by enhancing the predictability of project outcomes and timeframes for implementation.



Project preparation includes:

- Markets and Demand Analysis
 - Raw Materials and Supplies Study
 - Technical Analysis
 - Location and Site
 - Environment Impact Assessment (EIA)
 - Raw Materials and Supplies Study
 - Technology Selection
 - Financial and Economic Analysis
 - Organizational and Human Resource
- 



Markets and Demand Analysis

Market and demand analysis is conducted to know about the market share that the proposed project will enjoy and aggregate demand for the product or service.

This analysis helps management determine if they can successfully enter a market and generate enough profits to advance their business operations.



Objectives of Market and Demand Analysis

- ☛ To identify potential consumers or buyers.
- ☛ To indicate the total current demand.
- ☛ To identify the pattern of demand temporally as well as geographically.
- ☛ To know customers' willing to pay for the proposed services or product.
- ☛ To identify appropriate channels of distribution for the product.



The analysis cover mainly:

- Aggregate demand for the proposed product
 - Market share of the project under appraisal (The share of unfulfilled demand)
 - Potential Buyers
 - Market price of product
 - Level of competition (Analysis of competitors, their strength and weakness)
 - Distribution channel/mechanism
- 

Source for Market related Information

Collection of Secondary Data

Sources of secondary data: National Sample Survey report, census, plan report, statistical abstracts, economic surveys industry potential reports by various agencies & website of industry associations, web-sites of relevant ministry, other information available in public domain, etc.

Conducting a market survey



The **key steps** in market and demand analysis are as follows:

- Situational analysis
 - Gathering secondary data
 - Conduct of market survey (primary data)
 - Market classification
 - Demand forecasting
 - Market planning
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Raw Materials and Supplies Study

- Concerned with **defining** the materials and supplies required, **specifying** their properties, and setting up their **supply program**.
- **Basis** for the selection of materials and input is the ***demand analysis*** and the ***production program*** and finally ***plant capacity***.

Supply Marketing and Supply program

Supply Marketing

Cost minimization, risk minimization (reliability of suppliers) and the relations with supplier.

Cost Minimization

Input costs can be reduced by selecting appropriate suppliers and by choosing a proper volume and frequency of the orders.



Risk minimization (reliability of suppliers)

Reliability with regards **quantities**, **qualities**, **deadlines** and **prices** is significant for the entire manufacturing process.

Relations with suppliers

- ☐ Establishing **smooth and productive relations** with the suppliers
 - ☐ Building up maximum **bargaining power**
- 

Supply Program

A *supply program* should deal with

- ▶ Identification of supplying sources and supplies
- ▶ Quantities and qualities
- ▶ Agreement and regulations
- ▶ Consignments
- ▶ Means of transport
- ▶ Storage
- ▶ Risk assessment



Costs of Raw Materials and Supplies

- unit costs,
 - annual costs and
 - overhead costs
- 

Technical Analysis

The technical analysis is used to establish whether or not a project is **technically feasible**.

The technical feasibility analysis should consider various aspects of a project as follows:

- Project Design
- Selection of the production process
- Specification of equipments
- Buildings
- Engineering works
- Choice of technology
- Standards
- Plant layout
- Project location and site
- Production Plant Capacity
- Size of project



Location, Site, and Environmental Assessment

Determine the location and the site suitable for a project.

Location and site are often used synonymously but must be distinguished.

Location refers to a fairly **broad area** (a region, a city, an industrial zone) while site refers to a **specific piece of land** where the project would be set up.





Location and Site Analysis

Location and site analysis determine the location and the site **suitable** for a project.

Location and site are often used synonymously but must be distinguished.





Location is fairly **broad area** like a city, town, an industrial zone, region, country.

Site is a specific **piece of land** where the project would be set up.



Requirements to be identified in location analysis:

- ❑ Natural environment
- ❑ Geophysical conditions
- ❑ Raw materials availability
- ❑ Availability of labor (skilled and unskilled)
- ❑ Incentives and restriction
- ❑ Government plans and policies
- ❑ Infrastructure services
- ❑ Urbanization and literacy



In the selection of location, you should consider:

- The **availability** of raw materials and inputs.
 - The **proximity** to raw materials & centers of consumption.
 - The **existence** of basic infrastructure facilities
- 



The **optimal location** is where the total cost (raw material **transportation cost** plus **production cost** plus **distribution cost** for final product) is **minimized**.

A **resource based** project like a cement plant or a steel mill should be located **close to the source** of basic material (limestone in the case of cement plant and iron- ore in the case of a steel plant).

A project **based on imported** material may be located **near a port**.



Site Selection

Main Considerations

The following factors determine the selection of the final site

- Cost of land
- Site preparation cost
- Cost of utility lines extension
- Environmental considerations
- Size and shape of the available area
- Suitability for future extension
- Proximity of centers of consumption (market orientation)

- 
- Infrastructure facilities (transport network, houses, power supply etc.)
 - Availability of labor in the area
 - Socio – economic factors
 - Distance to seaport (import and export)
- 



The **cost** estimates at the site are

- Acquisition of land
 - Taxes
 - Legal expenses
 - Site preparation and development.
- 

Environmental Impact Assessment (EIA)

Environment includes the conditions under which any individuals or thing exists, live or develop.

Environmental Impact Assessment is concerned with the identification, prediction and evaluation of the **impact** of the proposed project aimed at **eliminating or minimizing** damaging impacts and **optimizing** beneficial impacts.

EIA will lead to a **decision** to **accept, modify** or **reject** a project.

Purpose of the Assessment

- ☛ To identify and assess any potentially adverse **environmental effects** of a new development.
- ☛ To make sure that the adverse impacts could be **avoided or reduced**.
- ☛ To ensure that environmental consequences were **taken into account** during planning, designing and decision Making process.
- ☛ To influence how it is subsequently **managed** during its implementation.



Stages of EIA process

- **Identifying** of various potential impacts of the project on the environment
 - **Predicting** of the extent of the environmental changes.
 - **Assessing** of whether or not the identified and predicted changes are of any environmental significance.
 - **Planning** of mitigation measures or alternatives that could reduce the project's environmental impacts.
- 



EIA report normally include the following information:

- ☛ The impact the project would have on the physical environment.
 - ☛ Any possible pollution of the soil, of waters of all kinds such as surface, underground, costal and of the atmosphere.
 - ☛ The impact of the project on wildlife, the natural habitat and all other ecological factors.
 - ☛ The project's likely influence on the qualities of life of the local populations.
 - ☛ Any influence the project may have on existing industry and employment.
 - ☛ Any need that may result for new or improved infrastructure such as utilities, transport, housing, school recreational amenities etc.
- 



Technology Selection

Determination of the appropriateness of technology for the project.

The required machinery and equipment must be determined in relation to the technology to be utilized, the local condition and human capabilities.





Factors should be taken into consideration are as follows:

- ✔ The projects ability to utilize local resources.
 - ✔ Its wider application. Consider easily adaptable technology.
 - ✔ Sustainability. There should be adequate services or maintenance facilities for the selected technology.
 - ✔ Reliability.
 - ✔ Implementers and beneficiaries ability to operate the chosen technology.
 - ✔ Cost-effectiveness of the chosen technology in comparison to the alternatives.
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Organizational and Human Resource

In general there are three basic types of organizational arrangements for implementing a project as appropriate

1. Functional organization. This involves the use of existing structural lines within an organization. This structure is suitable for small projects where co-ordination across functional lines is liable to be less problematic.
2. Projectised organization. This involves the creation of separate and self-contained organization with special goal of project implementation. This form of structure is often set up for large and complex projects. It will be banded when the project is completed.
3. Matrix organization. This form of structure integrate both the functional and projectised organization.



Human Resources

Identify the **requirements** (with sufficient skills and experience) and assess the **availability** of human resources as well as the training needs.

The **successful** operation of a project will ultimately depend on the **skill**, **experience** and **productivity** of workers and management.



Financial and Economic Analysis

Financial Analysis

The financial analysis is one of the analyses to be conducted in a feasibility study and is normally undertaken after the market and technical analysis.

The objective of the financial analysis is to determine the financial viability of the project.

This analysis determines if the project is a **profitable investment**, especially to the individuals or agencies undertaking the investment. If it is not profitable, then it will be financially prudent to consider alternative investments to maximize the use of capital and other resources.



Financial viability can be judged on the following parameters:

- ✓ Total estimated cost of the project
 - ✓ Sources of project finance
 - ✓ Projection of Sales and Cost of Production
 - ✓ Projected balance sheet, profitability, and cash flow.
- 

A) Cost of project:

Conceptually, cost of project represents the total of all items of outlays associated with a project. It's the sum of the outlays on the following:

- Land and site development costs
- Building and civil works costs
- Plant and machinery costs
- Technical know-how and engineering fees
- Expenses on foreign technicians and training of technicians abroad
- Miscellaneous fixed asset
- Preliminary and capital issue expenses
- Working Capital
- Pre-operative expenses e.t.c.

B) Sources of project finance:

- ✔ **Share Capital:** there are two types of share capital, namely equity capital and preference capital. *Equity capital* represents the contribution made by the owners of business, the equity shareholders, who enjoy the reward and bear the risks of ownership. *Preference capital* represents the contribution made by preference shareholders and the dividend paid on it is generally fixed.
- ✔ **Term loan:** are provided by financial institutions and commercial banks.
- ✔ **Debenture capital:** debentures are instruments for raising debt capital.
- ✔ **Deferred credit:** many a time the suppliers of the plant and machinery offer a deferred credit facility under which payment for the purchase of the plant and machinery can be made over a period of time(Hire purchase).
- ✔ **Incentive source:** the government and its agencies may provide financial support as an incentive to for setting up industrial units in certain locations.
- ✔ **Other sources**

C) Projection of Sales and Cost of Production

i) Projected Sales

Typically, the starting point for profitability projections is the forecast of sales revenue.

ii) Cost of production

The major components of cost of production are:

- Material cost which comprises the cost of raw materials, chemicals, components, and consumable stores required for production.
- Labor cost which is the cost of all the manpower employed in the factory.
- Utility cost consists of power, water and fuel determined based on norms or consumption standards in the industry.
- Factory overhead cost: the expenses on repairs and maintenance, rent, taxes, insurance on factory assets, and so on are collectively referred to as factory overheads.



D) Balance Sheet, Profitability, Cash Flows and Projections

Given the estimates of sales revenue and cost of production as discussed above, at least it is fair to provide projected Balance Sheets, Income Statements and cash flow projections for the life of the project.





Financial evaluation

- Payback period of the project (PBP)
 - Accounting rate of return (ARR)
 - Net present value (NPV)
 - Internal rate of return (IRR)
 - Profitability index (PI)
- 

Economic Analysis

Cost and benefits to the society or to the nation due to the proposed project are considered in the economic feasibility test.

Thus, economic feasibility is also called **Social Cost Benefit Analysis (SCBA)**.

Cost-Benefit analysis (CBA) deals with financial or profitability analysis whereas Social cost-benefit analysis (SCBA) deals with the benefit the society will get or the cost the society has to bear.



Tax revenue, generation of employment, foreign earnings, savings and investments, **distribution of income** and such other factors, differentiate economic feasibility from financial viability.

Economic aspect addresses the project's impact and its worthwhileness from the viewpoint of the whole society.

The government and government agencies calculate the economic indicators of a project before **permitting the project** or **financing it**.





Political Feasibility

Political viability of the area/country where the project will be established is an important facets of feasibility study.

There are two types of politics to be considered: Internal and external.

Internal politics this is related with internal political stability.

The second type of politics is external politics related to regional political stability.



Stakeholders Analysis

Stakeholder analysis is the process of identifying the people, groups, communities and institutions who/which are liable to be affected in some way by a proposed project.

Stakeholder is a person or group of people who have a vested **interest** in the **success** of a project and the **environment** in which the project operates.

Typical Stakeholders

- Project Sponsor
- Project team
- Supplier
- End User
- Community
- Steering Committee
- Customer
- Contractor
- Environmental Agency
- Functional managers



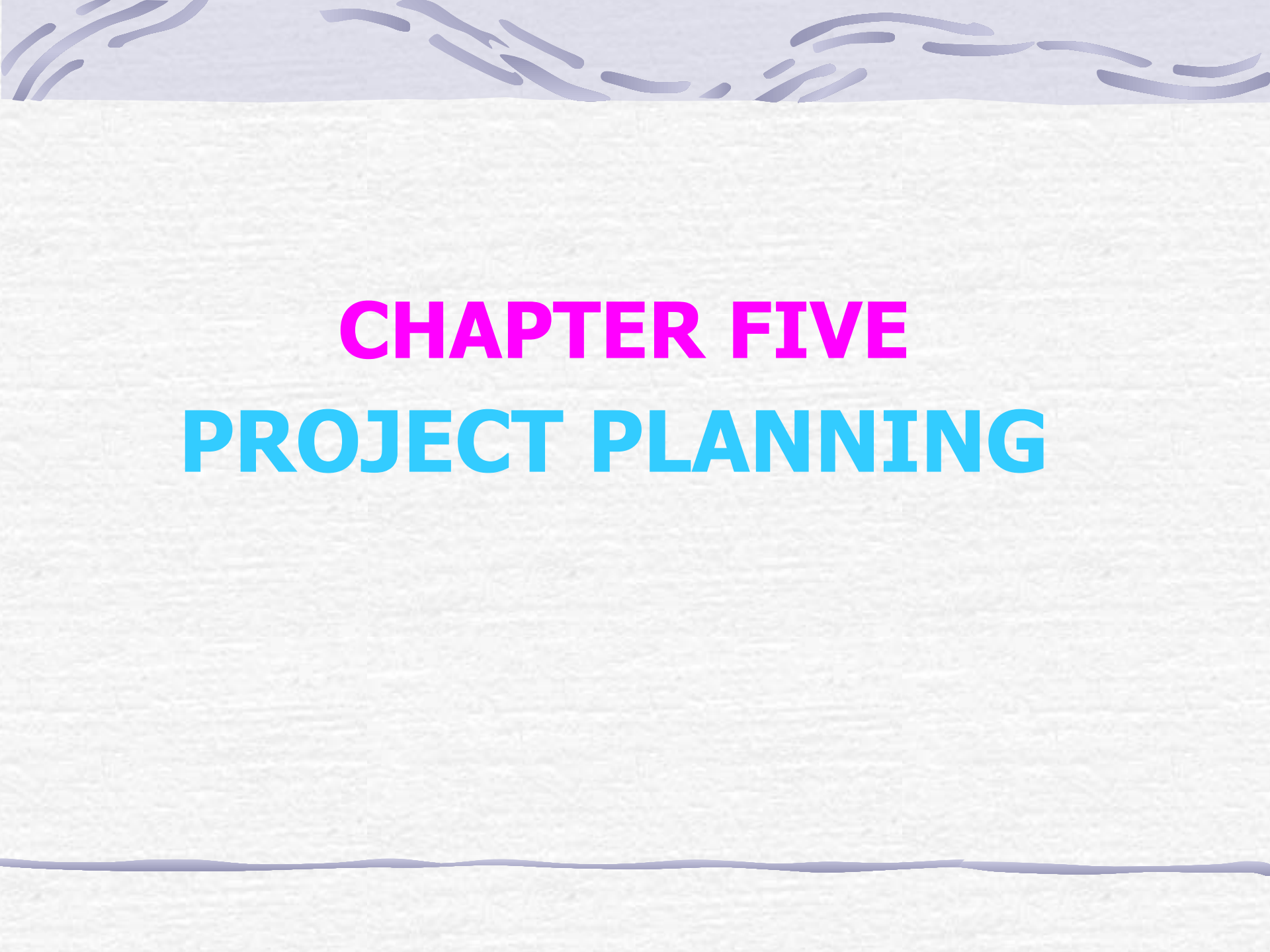
Once these stakeholders have been identified it is important to establish key information about them such as their **interests**, potential **impact** (negative or positive) on the project, the relative **importance** of their interest.

The **influence** and **importance** of the stakeholders in terms of achieving project objective should be considered.

Key stakeholders are those who either have **a significant influence** over a project, or who are **important** in terms of meeting project objectives, or **both**.

When dealing with **participation** it is important to determine their involvement in terms of inform, consult, partnership, or control.





CHAPTER FIVE

PROJECT PLANNING



Planning is an essential part of project management.

Planning determines **what** need to be done, **who** will do it, **how long** it will take, and **how much** it will cost.

It is important that the people who will be **involved in performing the work** are also **involved** in planning the work. **Participation** builds commitment.





Planning Phase Involves

- Project goals and objectives
 - Project Scope
 - Work breakdown structure
 - Network diagramming
 - Scheduling (Estimate time)
 - Resource requirements (Distribute resources)
 - Cost Estimate
- 



Project Goals and Objectives

The first step in the planning process is to define the project goals and objectives. The goals and objectives must be clearly defined and agreed upon by the customer and the organization or contractor that will perform the project.



Project scope

The project scope is the **framework** of a project. The scope determines the "**boundaries**" of the project and defines exactly what the project will achieve.

The project scope is the basis of the project, and all other planning activities will revolve around the project scope.

The project scope is usually documented as a **scope statement**.

Ensure that the **project includes all of the work required**, and only the work required, to complete the project successfully.

A project's scope should be based on the needs and resources.



Project scope statement consists

- Project objective
 - Deliverables
 - Milestones
 - Exclusions
 - Technical requirements
 - Reviews with customer
- 



Deliverables

A deliverable is any **product**, **service** or **result** that must be completed in order to finish a project.

A deliverable is a tangible or intangible good or service produced as a result of a project that is intended to be delivered to a customer (either internal or external).

The **expected outputs** to be provided over the life of the project.

Something produced or provided as a result of a process.



Examples of deliverables

- Report (Status reports, Service report Strategic report Progress report)
- Proposal
- Design (Design drawings, Design documents Design review, a new or improved design)
- Completed product (house, building, bridge, etc.)
- Document (Tender document, instruction manual)
- Software products/programs
- Marketing study
- Training programs
- Meetings, reviews
- Product quality enhancement
- New equipment or tools
- Better customer service
- Web site/page
- New knowledge or experience
- Improved filing and organization systems
- Improved process



Milestones

A milestone is a **significant event** in a project that occurs at **a point in time**.

A milestone is a reference point that marks an **important event** in a project task or group of tasks.

Milestones have no duration themselves. They have zero duration because they **symbolize an achievement**, or a point of time in a project.

Milestones can be used to symbolize anything that has **started** or **finished**.

The **most significant** of all project management milestones, of course, is the one that marks the **completion of a project**.





They help project team members and other stakeholders stay informed about the project status.

They can do more than just show progress—they can help you communicate what's happening with your project.

They are also useful for celebrating project achievements.





Examples:

- ☐ The start of significant phases of a project
 - ☐ The end of significant phases of a project (completing a phase of the project)
 - ☐ Approval (the initial approval/Project approval that allows the project to move forward, design Approval final approval).
 - ☐ Contract signed
 - ☐ When an important decision is being made
 - ☐ When building a house (The floors will be finished on Monday, The roof will be completed on November 1st, The gas installation will be connected at the end of the month)
 - ☐ Requirements review (When projects involve a lengthy process)
- 

Exclusions

Exclusions further define the **boundary of the project** by stating **what is not included**.

Examples:

- Data will be collected by the client, not the contractor.
- A house will be built, but no landscaping or security devices added.
- Software will be installed, but no training given.

Work Breakdown Structure

After defining the objectives of the project and scope, the next step is to **break the project down** into manageable pieces in a work breakdown structure.

This requires developing a list of all the activities.

A Work Breakdown Structure (WBS) is a **hierarchical tree structure** of tasks (from general to specific) that need to be performed to complete a project.

WBS is the **basis** for time estimating, resource allocation, and cost estimating.



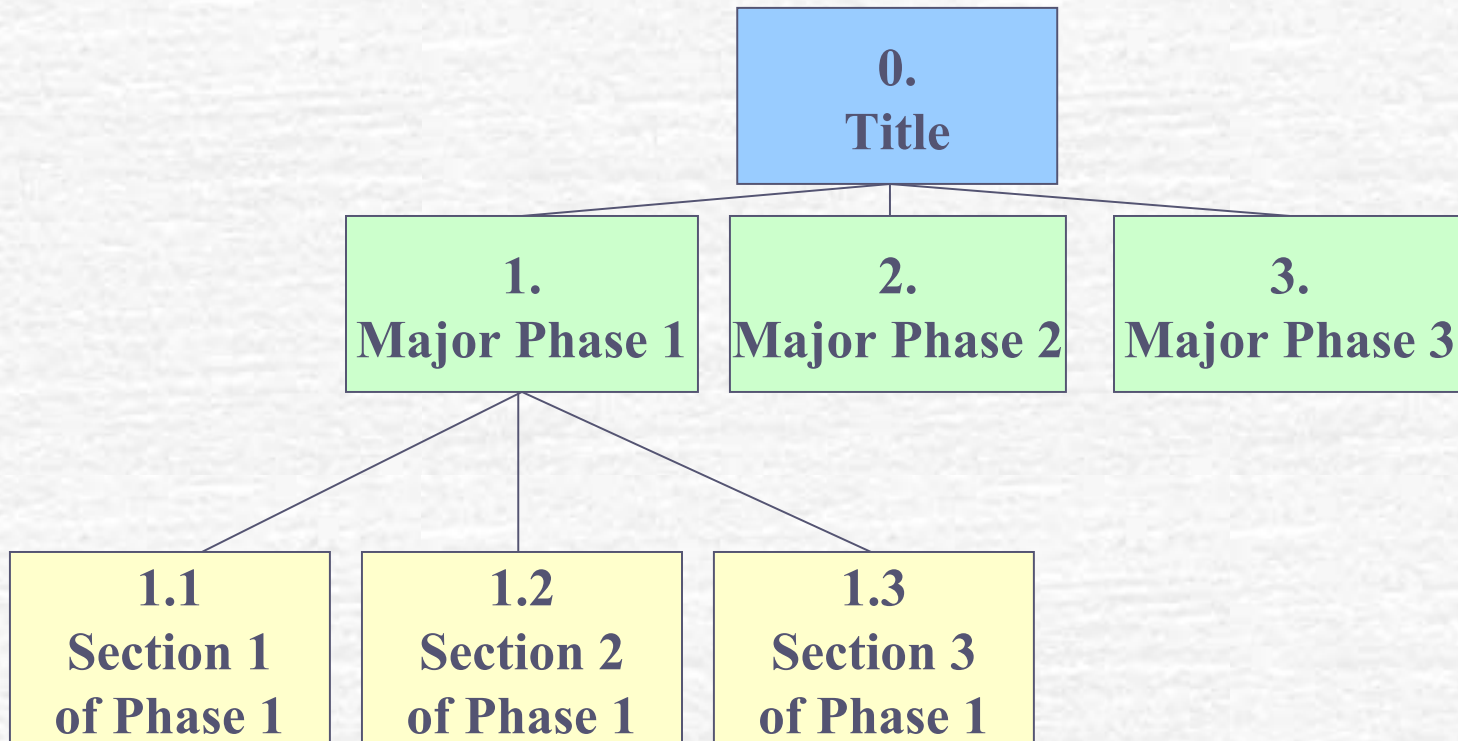
The **entire project team** should be **involved** in developing the work breakdown structure.

When completed, you should **review** the work breakdown with the concerned stakeholders. This is to ensure that it is complete and that it addresses their specific concerns.



WBS- Formats

1. Organization Chart Method



2. Outline Method

- 0. TITLE**
- 1. MAJOR PHASE 1**
 - 1.1 S1 OF PHASE1**
 - 1.2 S2 OF PHASE1**
 - 1.3 S3 OF PHASE1**
- 2. MAJOR PHASE 2**
 - ...**
 - etc.**

Example: WBS –Equipment System

WBS	Task Name
1	Equipment System
1.1	Initiation (Definition)
1.1.1	Develop SOR
1.1.2	Write Project Charter
1.1.3	Get Project Approval
1.1.4	<i>Project Approved</i>
1.2	Plan (Development)
1.2.1	Develop System Specs(for ID)
1.2.2	Develop System Specs(for PPA)
1.2.3	<i>Preliminary Project Approval</i>
1.2.4	Write SOW
1.2.5	Write RFP
1.2.6	Receive Proposals
1.2.7	Review Proposals
1.2.8	Select Bid
1.2.9	Contract Approval
1.2.10	<i>Contract Awarded</i>
1.2.11	Form Project Team
1.2.12	Project Team Training
1.2.13	Develop Project Implementation Plan

1.2.14	Plan Systems Engineering
1.2.15	Plan ILS/Supportability
1.3	Execution (Implement)
1.3.1	Accept Deliverable
1.3.2	Do Testing
1.3.3	Re-accept as Necessary
1.3.4	Do Fielding
1.3.5	Provide User Training
1.3.6	Write Project Reports
1.3.7	<i>Implementation Completed</i>
1.4	Close Out
1.4.1	Do Post Project Review
1.4.2	Write Close Out Report
1.4.3	Transfer to In-Service Support
1.5	<i>Project Completed</i>

Project Scheduling

Scheduling is determining when the tasks can and must be **started**; when they must be **completed**; which tasks are **critical** to the timely completion of the project; and which tasks have **slack** and how much.

Scheduling includes:

- Define Activities
- Sequence Activities
- Estimate Activity Durations
- Develop Schedule

Scheduling Tools

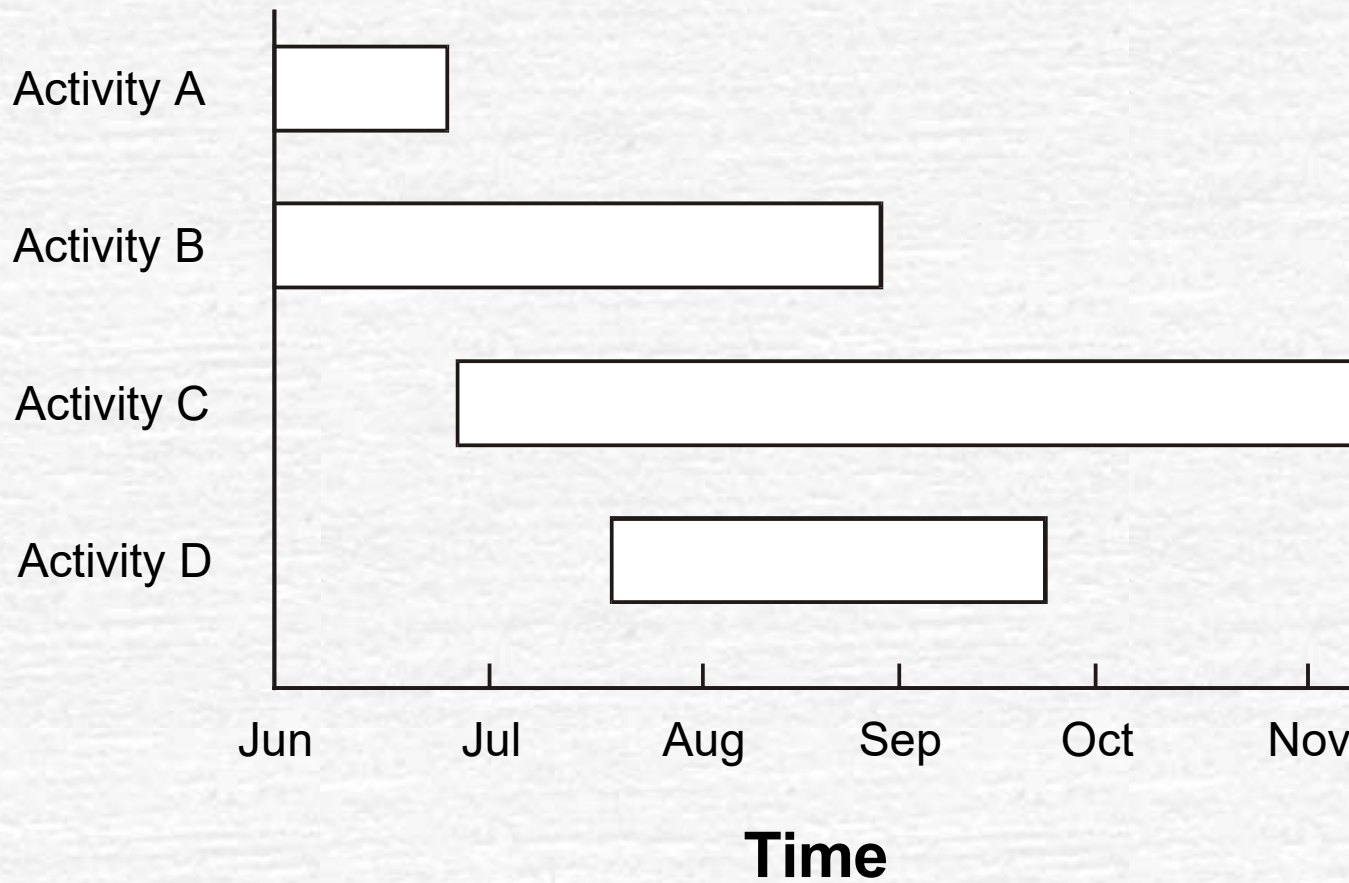
✓ Gantt Chart (Bar chart)

Gantt Chart is a popular type of bar chart that illustrates a project schedule.

Gantt chart is a **visual representation** of your project schedule.

A Gantt chart shows all of the **tasks (activities)** that need to be done, the **amount of time** each task is expected to take, the **time frames** in which individual tasks are to be completed, and the **relationship** between various tasks.

It helps get the work done on time.



There are many other acceptable ways to display project information on a bar chart.

WEEKS: 1 2 3 4 5 6 7 8 9 10 11 12 13

WBS 1 Summary Element 1

WBS 1.1 Activity A

WBS 1.2 Activity B

WBS 1.3 Activity C

WBS 1.4 Activity D

WBS 2 Summary Element 2

WBS 2.1 Activity E

WBS 2.2 Activity F

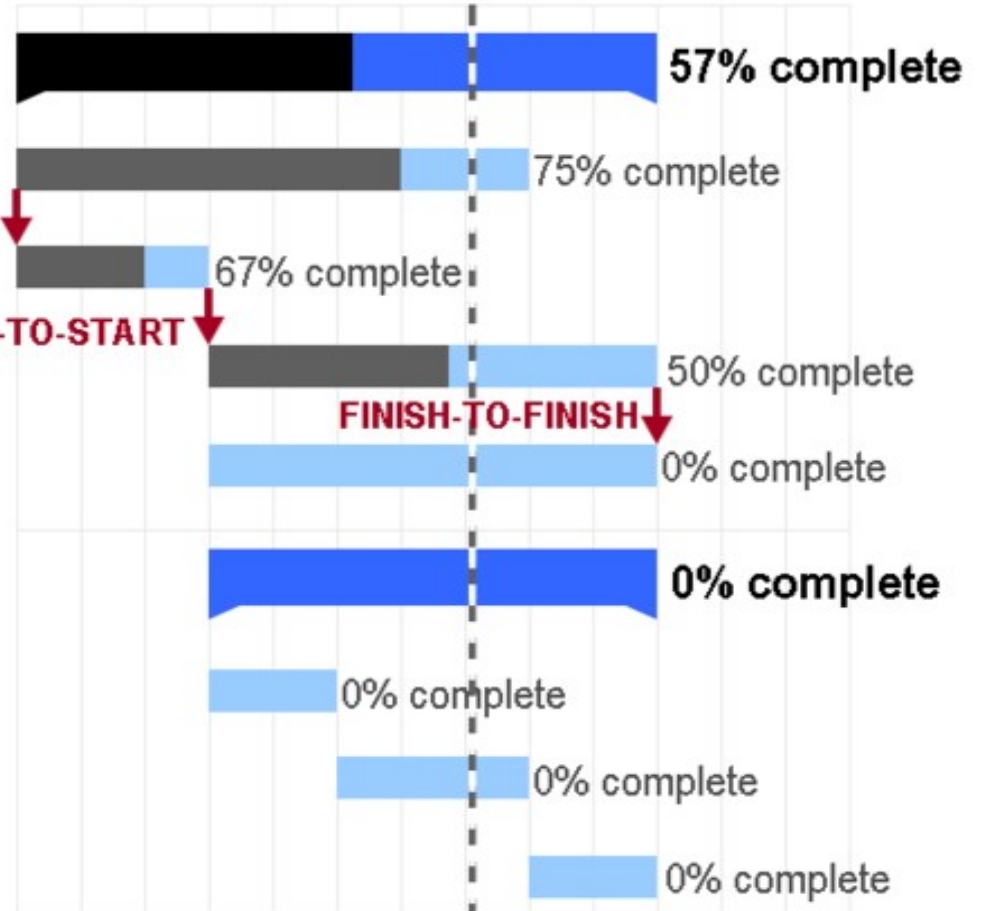
WBS 2.3 Activity G

START-TO-START

FINISH-TO-START

FINISH-TO-FINISH

TODAY





✓ Network Diagram

Network diagram is a **graphical flow** of the activities that must be accomplished to complete the project.

It shows the planned **sequence** of activities, **time requirements**, and **interrelationships**.

Drawn from **left to right**.



Terminologies

Parallel activities. Two or more activities that occur at the same time. Also called concurrent or simultaneous activities.

Sequential activities. Two or more activities that occur one after the other. Also called consecutive activities.

Diverging Activities (Merge activity): Single predecessor with multiple successors. This is an activity that has more than one activity immediately preceding it (more than one dependency arrow flowing to it).

Converging Activities (Burst activity): Multiple predecessors with single successor. This activity has more than one activity immediately following it (more than one dependency arrow flowing from it).

Critical path: It means the path(s) with the longest duration through the network; if an activity on the path is delayed, the project is delayed the same amount of time.

Event: This term is used to represent a point in time when an activity is started or completed. It does not consume time.

Formats of network diagram

Activity in the box (AIB)

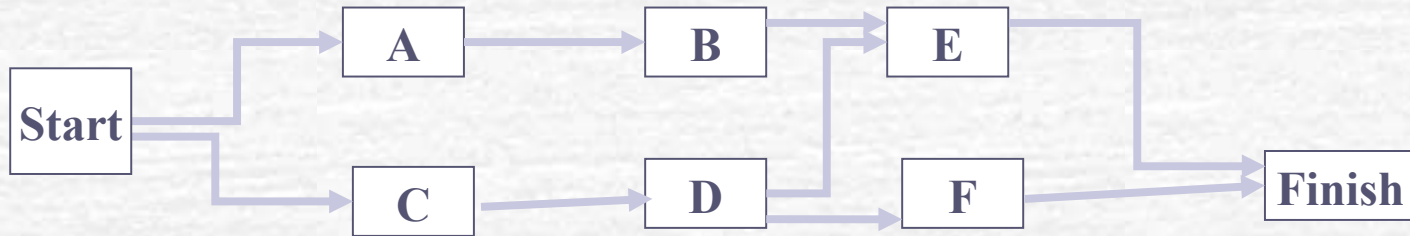
Each activity is represented by a box in the network diagram, and the description of the activity written in the box.

Activity on the arrow (AOA)

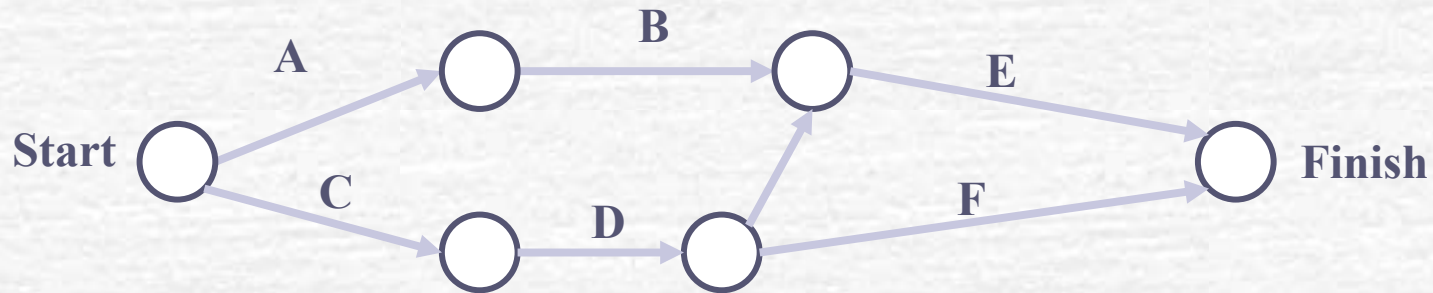
In the AOA, an activity is represented by an arrow in the net diagram, and the activity description is written above the arrow.

Network Diagrams

A, B, C, D, E, F are activities



AIB Format

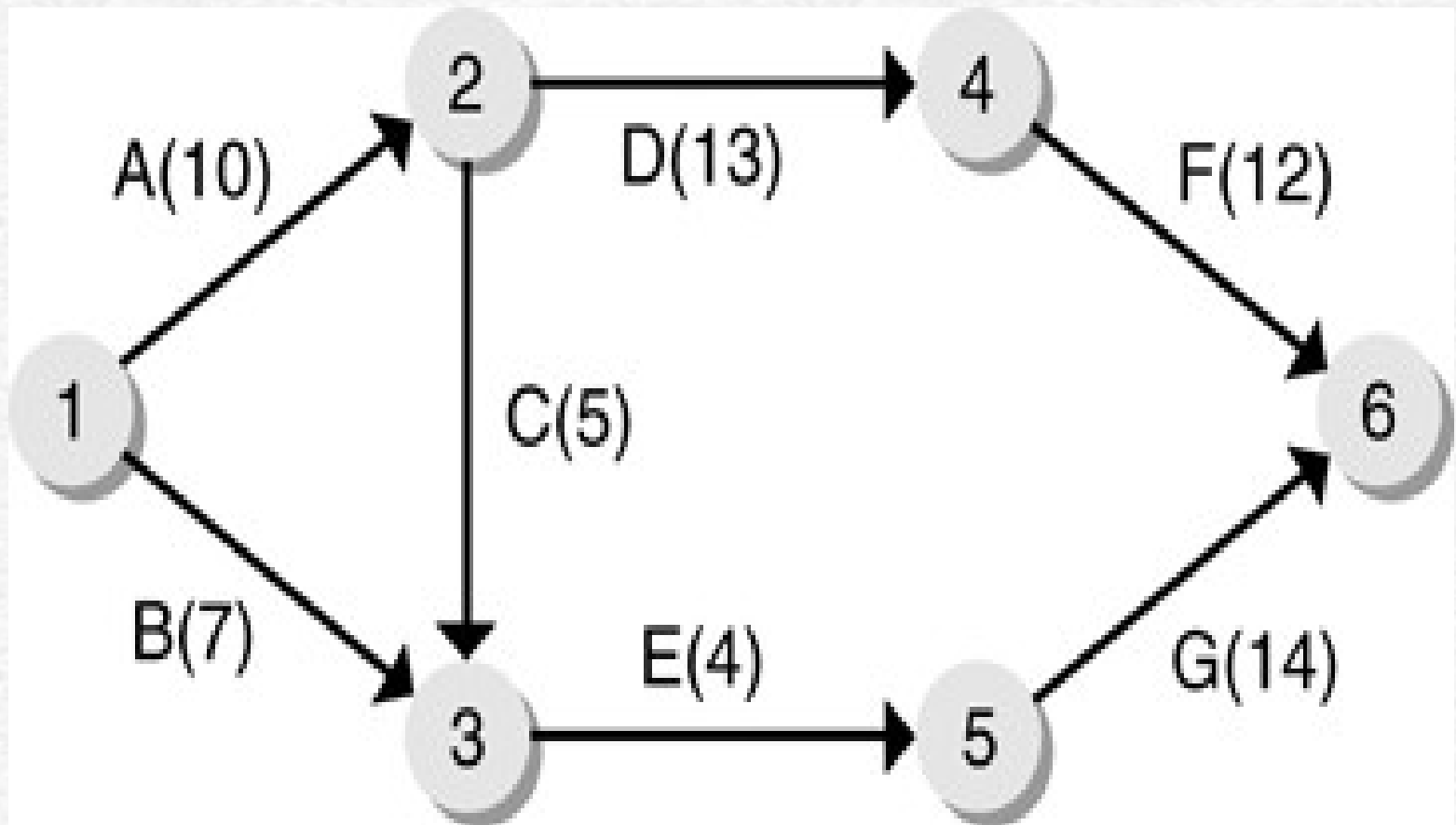


AOA Format

Example

Activity	Time(days)	Preceded By
A	10	--
B	7	--
C	5	A
D	13	A
E	4	B,C
F	12	D
G	14	E

Network Diagram





Time Estimation

Estimating of time is the process of **forecasting** or **approximating** the time of completing a project and an activity.

Estimating processes are frequently classified as **top-down** and **bottom-up**. Top-down estimates are usually done by senior management.

Bottom-up estimates are typically performed by the people who are doing the work. Their estimates are based on estimates of elements found in the work breakdown structure.

Accuracy is improved with greater effort. Inaccurate estimates lead to **false expectations** and **customer dissatisfaction**.

Increase estimates if workers will be **distracted by other responsibilities** or interrupted by other projects. It takes the worker more time to complete his or her work because time is wasted shifting between projects.

Increase the estimates when using **less-skilled workers** or workers with whom you have little experience.





Duration Estimate

Duration estimate is an estimate of **how long each activity will take**, from the time it is started until the time it is finished. This duration estimate for each activity must be the time for the work to be done plus any associated waiting time.

Start and Finish Times

It is necessary to select an estimated **start time** and a required **finish (completion) time** for **the overall project**. Also estimate start and finish times **for each activity**.





Determine earliest and latest times

Earliest start time for activity (ES)- How soon can the activity start?

Latest start time for the activity (LS)- How late can the activity start?

Earliest finish time for the activity(EF)- How soon can the activity finish?

Latest finish time for the activity (LF)- How late can the activity finish?



Forward Pass—Earliest Times (ES and EF)

The forward pass requires that you remember the following when computing early activity times:

- You **add activity times** (duration) along each path in the network to compute early finish (**$ES + DUR = EF$**).
- You **carry** the early finish (EF) **to the next activity** where it becomes its early start (ES).
- The next succeeding activity is a **merge activity**. In this case you select the largest early finish number (EF) of all its immediate predecessor activities.

Backward Pass- Latest Times (LS and LF)

The backward pass is similar to the forward pass; you need to remember three things:

- You **subtract activity times** along each path starting with the project end activity to find the late start (**$LF - DUR = LS$**).
- You **carry the LS** to the next preceding activity to establish its LF, unless.
- The next preceding activity is a **burst activity**; in this case you select the **smallest LS** of all its immediate successor activities to establish its LF.

Determining Slack (or Float)

When the forward and backward passes have been computed, it is possible to determine which activities can be **delayed** by computing “slack” or “float.”

Total slack or float for an activity is simply the difference between the LS and ES or LF and EF

$$\text{Slack} = \text{LS} - \text{ES} \quad \text{or} \quad \text{Slack} = \text{LF} - \text{EF}$$

After slack for each activity is computed, the critical path(s) is (are) easily identified. The critical path can be identified as those activities that also have $\text{LF} = \text{EF}$ or a slack of zero ($\text{LF} - \text{EF} = 0$), or $\text{LS} - \text{ES} = 0$).

Schedule for a Project

	Activity	Responsi bility	Duration	Earliest		Latest		Total Slack
				Start	Finish	Start	Finish	
1	A	Jim	10	0	10	0	10	0
2	B	Jim	7	0	7	10	17	10
3	C	Susan	5	10	15	12	17	2
4	D	Abera	13	10	23	10	23	0
5	E	Kebe	4	15	19	17	21	2
6	F	Amare	12	23	35	23	35	0
7	G	Amare	14	19	33	21	35	2

When Expected Activity Times are Uncertain

Optimistic estimate- the longest time for each task (t_o), most likely- the mean value of the estimates of each task (t_m), and pessimistic estimate- the shortest time for each task (t_p) time estimate for each activity. t_e = Estimated time (PERT Estimating)

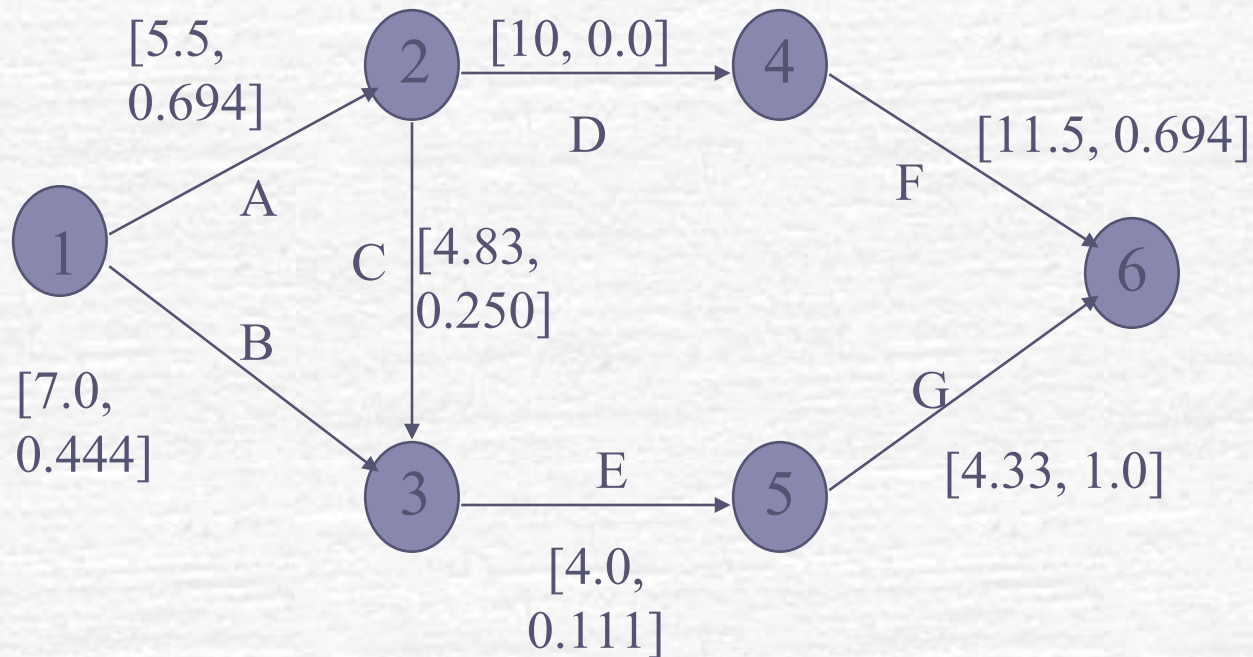
$$t_e = \frac{t_o + 4t_m + t_p}{6}$$

$$\sigma^2 = \left(\frac{t_p - t_o}{6} \right)^2$$

Example

Activity	Preceded By	t_o	t_m	t_p	t_e	σ^2
A	--	2	6	7	5.50	.694
B	--	5	7	9	7.00	.44
C	A	3	5	6	4.83	.250
D	A	10	10	10	10.0	0.000
E	B,C	3	4	5	4.0	.111
F	D	8	12	13	11.5	.694
G	E	2	4	8	4.33	1.000

Network Diagram with Expected Activity Times and Variances



Expected Completion Time and Variance of Path A-D-F

Expected completion time = $5.5 + 10 + 11.5 = 27$



Assigning Resources

The best project plan in the world cannot be accomplished without the right people, materials, and equipment at the right place at the right time.



Consider the following principles when assigning resources:

- Schedules are meaningless unless the **right resources are available** when the activity is scheduled to begin.
- If you **cannot get the right resources** at the right time, you may need to **replan**.
- **Do not assign the wrong person** to the job just because no one else is available at that time. Assign the most appropriate people to each activity.

Identify the skills required for each activity and recruit individuals who best meet the skill requirements. You don't want to assign an entry-level editor to do a job specified for a senior editor. Likewise, you usually would not assign a senior (and higher paid) editor to do a job that could be accomplished by an entry-level editor.

- **Assign scarce resources** to activities on the **critical path first**.
- 

- 
- **Obtain firm commitments** from team members, functional managers, and senior management.
 - **Too few people** on a project cannot solve the problems; **too many** people can create more problems than they solve.
 - It may be necessary to **increase the project** duration to get the right resources at the right times.
- 



Project Cost Estimates

The project cost estimate is allocated to the various **work packages** in the WBS.

The budget for each work package is distributed over the duration of the work package so that how much of its budget should have been **spent at any point in time**.





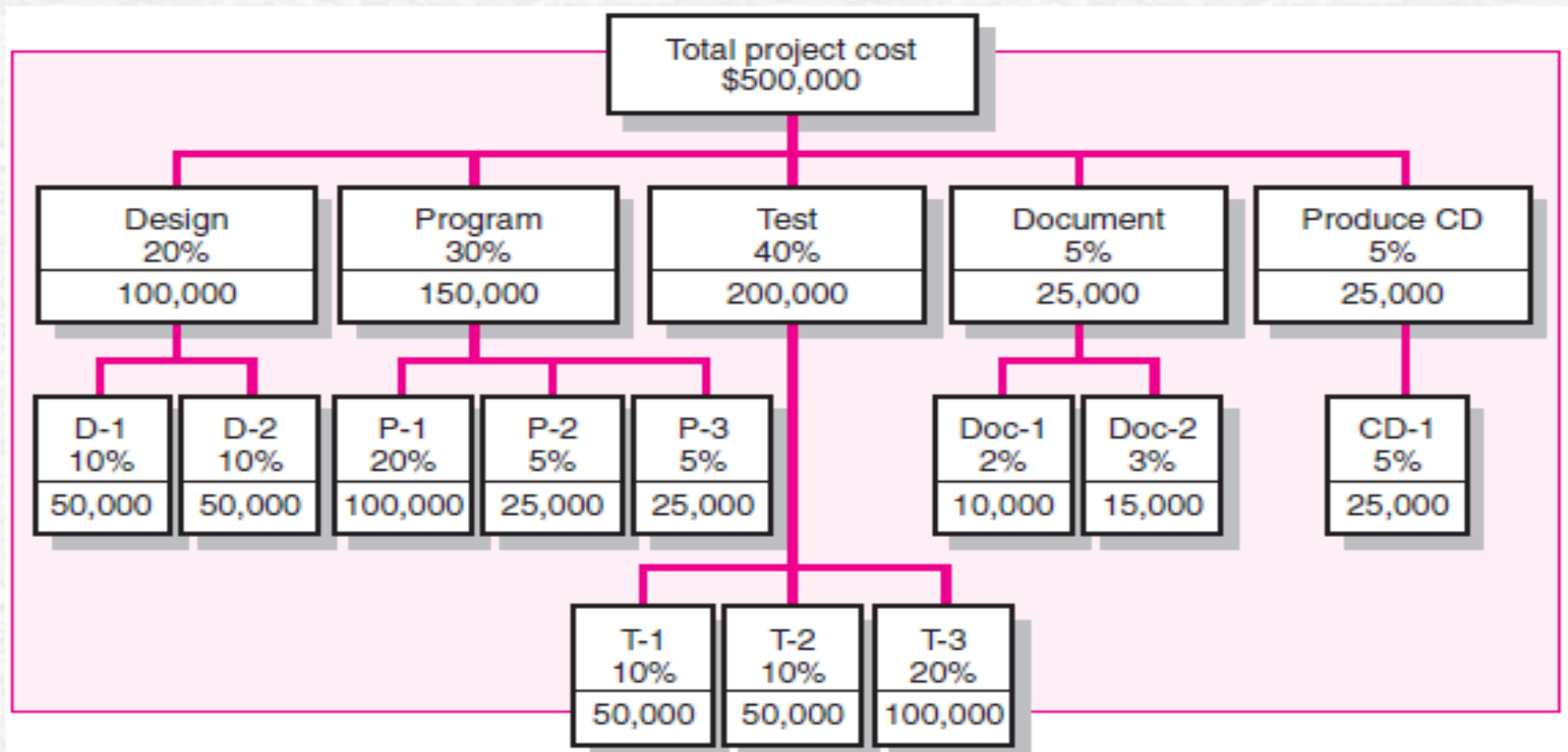
Allocating Total Budgeted Cost (TBC)

The amount allocated to each work package represents the TBC for **completing all the activities** associated with the work package.

When the budgets for all the work packages are **summed**, they **cannot exceed** the total budgeted cost of the project .



The Work Breakdown Structure with the total budgeted cost for each work package

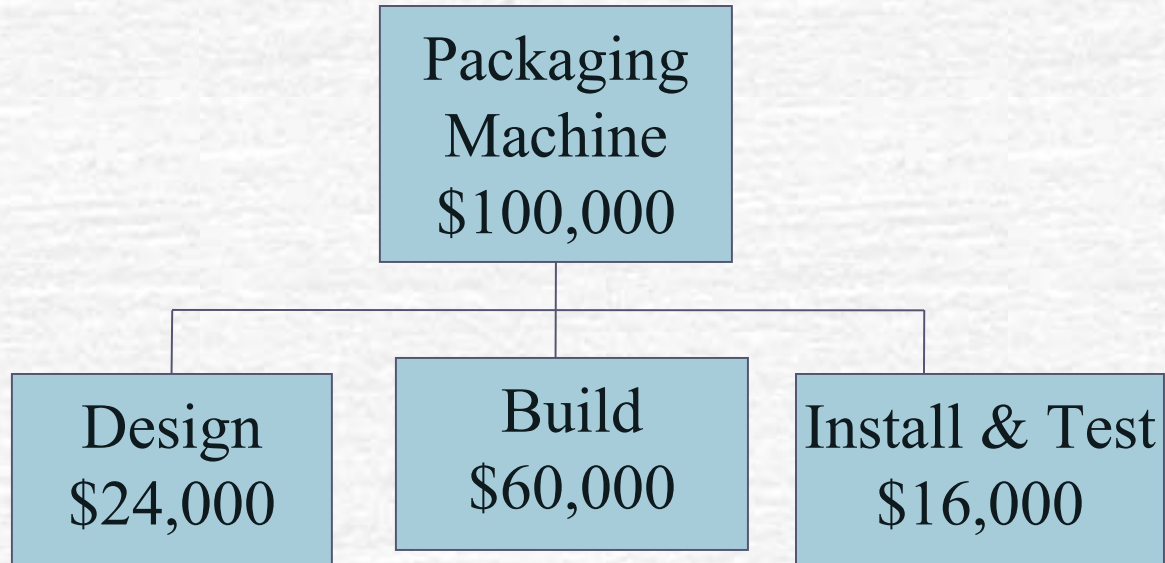


Developing Cumulative Budgeted Cost

Once a TBC has been established for each work package, the second step in the project budgeting process is to **distribute each TBC over the duration** of its working package. A cost is determined for each period based on when the activities that make up the work package are scheduled to be performed.

When the TBC for each work package is spread out by time period, it can be determined how much of the budget should have been spent at any point in time.

Example: WBS for packaging machine with TBC



Budgeted cost by period for the packaging machine project (In thousands)

Work package	TBC	Week											
		1	2	3	4	5	6	7	8	9	10	11	12
Design	24	4	4	8	8								
Build	60					8	8	12	12	10	10		
Install & Test	16											8	8
Total	100	4	4	8	8	8	8	12	12	10	10	8	8
Cumulative		4	8	16	24	32	40	52	64	74	84	92	100



Project Plan

The **end result** of your planning phase is a document called the **project plan**.

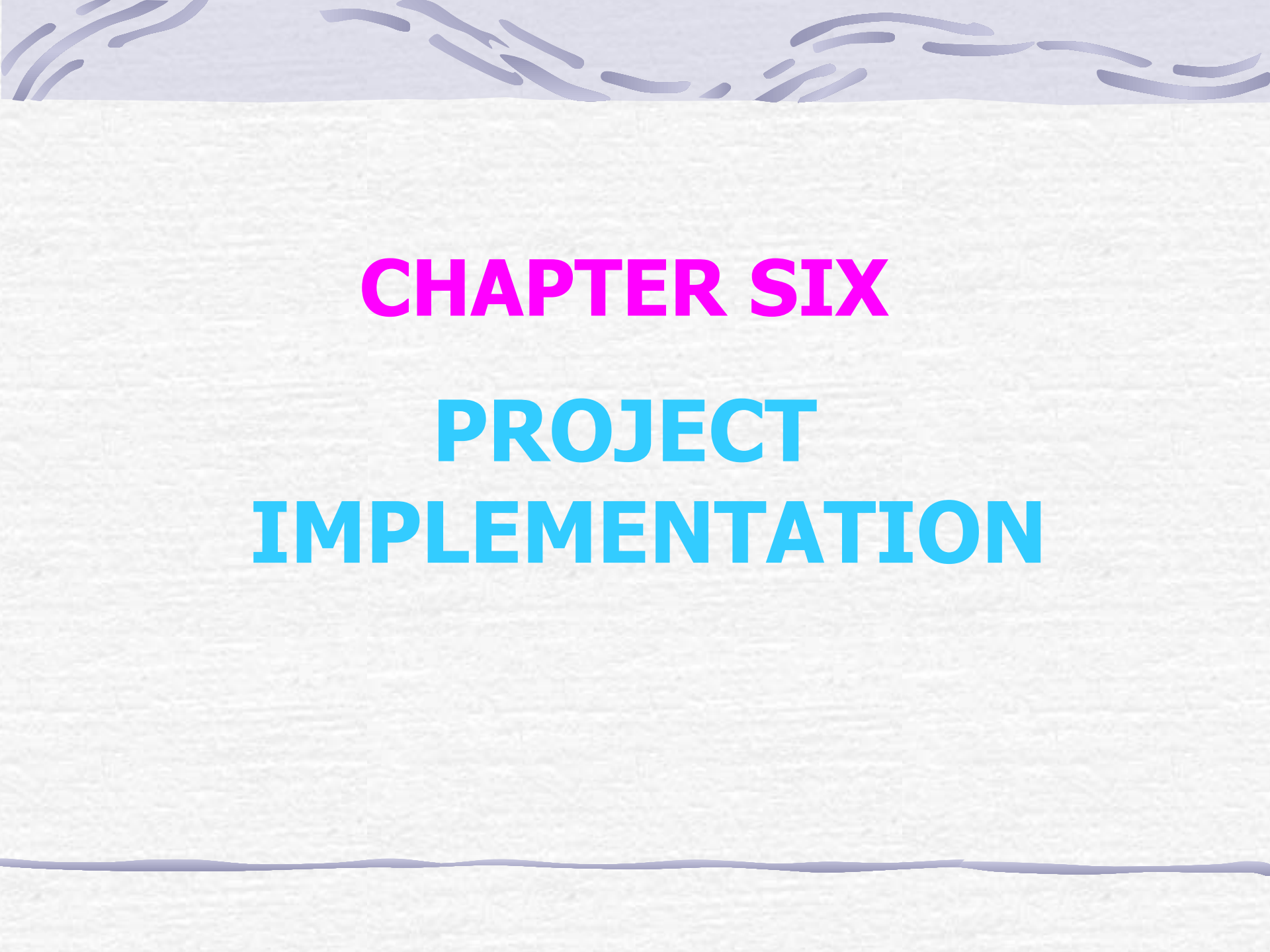
A project plan is "a formal, approved document used to **guide** both project **execution** and project **control**."

The project plan may be made up of lots of subsidiary plans. These include:

- A plan for managing the human resources.
- A plan for managing costs and the budgeting.
- A plan for procurement
- A plan for communications
- A risk plan for dealing with project risk
- A quality plan that specifies the quality targets for the project.

That's **a lot** of documentation. In reality, **it's rare** that you'll produce these **as individual documents**. What you need is a project plan that **talks** about the important **elements of each of these**.





CHAPTER SIX

PROJECT IMPLEMENTATION



After you have successfully passed the planning phase, it is time for real action- to start the project **implementation (or project execution)** phase, the third phase of the project management life cycle.

It is usually the **longest phase** in the project life cycle and it typically consumes the most energy and the most project resources.

The implementation phase turns **project plan into action**.

Project implementation is a process whereby project inputs are converted to project outputs.

The implementation phase is where project team **actually do the project work** to produce the deliverables.





Implementation tends to be complicated in practice by many unforeseen problems.

You should **not expect** the project to be implemented **exactly as planned**.

Flexibility is, therefore, required at this stage to enable the successful execution of the project.

No matter how good the original plan is, there will always be **some deviation** during implementation.

This should be anticipated, and the aim of project management is **to track this deviation**, make sure it **stays within the scope** of the project, and redirect activities to get back on track. Never allow activities to go outside the approved scope of the project.

Define margins and the scope for variation that can be tolerated to achieve objectives with the available resources.





Project implementation involves:

- ✓ Establish and manage the project team.
 - ✓ Manage other resources.
 - ✓ Performing the activities needed to meet the project objectives.
 - ✓ Contracting procurements.
 - ✓ Recommend changes and implement approved changes.
 - ✓ Hold meetings.
 - ✓ Communication.
 - ✓ Control project objectives
 - ✓ Reporting.
 - ✓ Manage risks.
 - ✓ Monitoring performance.
 - ✓ Give recognition and rewards to team members and keep them motivated.
- 

Communicating Project Plans

Be sure to **communicate** project information to all stakeholders: the project team, functional managers, senior management, and customers.

Present the appropriate level of detail to each group. Management may be interested in **summary-level** information whereas team members need much more **detail**.

Be sure to cover all areas of planning, including:

- Statement of the project objectives
- Work breakdown structure
- Logic network diagrams
- Schedules for individual activities and the project as a whole
- Charts showing estimated costs and projected cash flow



Managing Team

It is the people—the project manager and the project team—that are **critical** to accomplishing the project objective.

In project management, the **skills and knowledge** of the people involved are vital for producing the result.

Project Manager

The project manager is a key ingredient in the success of a project. It is the responsibility of the project manager to make sure that the **customer is satisfied** that the **work scope is completed** in a **quality** manner, within **budget**, and **on time**.

The project manager has primary responsibility for providing **leadership** in planning, organizing, and controlling the work effort to accomplish the project objective. In other words, the project manager provides the **leadership to the project team** to accomplish the project objective.





In addition to providing leadership, the manager should possess a set of skills that will both **inspire** the project team to succeed and **win** the confidence of the customer.

Effective project managers should have strong **leadership ability**, the ability to **develop people**, excellent **communication skills**, good interpersonal skills, the ability to **handle stress**, **problem solving** skills, and **time management** skills.





Project Team

Teamwork is the **cooperative effort** by members of a team to achieve that common goal.

A project team is more than a group of individuals assigned to work on one project. A project team is a group of **interdependent individuals** working **cooperatively** to achieve the project objective.

The effectiveness—or lack thereof—of the project team can **make the difference** between project **success** and project **failure**.

The project manager and the project team are the key to project success; project **success** requires an **effective project team**.

Therefore, you should be familiar with the development and maintenance of an effective project team is necessary.

Helping these individuals develop and grow into a cohesive, effective team takes effort on the part of the project manager and each member of the project team.



Communication

Communication in a project takes place between the project team and the customer, among the project team members, and between the project team and its upper management.

Communication may involve two people or a group of people. It can be **oral** or **written**. It can be face to face or involve some medium, such as phone, voice mail, e-mail, letters, memos, videoconferencing, or groupware.

It can be **formal**, such as a report or a presentation at a meeting, or **informal**, such as an e-mail message.

Effective and **frequent communication** is crucial for keeping the project moving, identifying potential problems, soliciting suggestions to improve project performance, keeping abreast of customer satisfaction, and avoiding surprises.



Project managers must be **good communicators**. They need to **communicate regularly** with the project team, as well as with any subcontractors, the customer, and their own company's upper management.

Good communication is especially important **early** in the project to build a **good working relationship** with the project team and to **establish clear expectations** with the customer.

Effective project managers **communicate** and **share information** in a variety of ways. They have **meetings** and informal conversations with the project team, the customer, and the company's upper management. They also provide **written reports** to the customer and upper management.

All these tasks require that the project manager have good **oral** and **written** communication skills.





The project manager should create an atmosphere that fosters **timely** and **open communication** without any fear of reprisal, and he or she must **accept differing viewpoints**.

It's important for the project manager to provide timely **feedback** to the team and customer. Both the good news and the bad news should be shared promptly. For the project team to be effective, members need to have **up-to-date information**—especially customer feedback that may necessitate changes to the project work scope, budget, or schedule.





Personal Communication

Effective and frequent **personal communication** is crucial to keep the project moving, identify potential problems, solicit suggestions for improving project performance, keep abreast of whether the customer is satisfied, and avoid surprises.

Listening

We learn more by **listening than by talking**. Therefore, good project managers spend more time listening than talking. They **don't dominate** a conversation.

The **heart** of communication is not words, but **understanding**—not only to be understood, but also to understand. Half of making communication effective is listening. Failure to listen can cause a breakdown in communication.

Listening is more than just letting the other person talk. It must be an active, not a passive, process. **Active listening** increases understanding and reduces conflict.





Reporting

Reports should be designed to communicate exactly what needs to be communicated.

The report includes information about activities carried out, outputs delivered and expenditure incurred, and problems that have been encountered. It makes sense to focus reports on time, cost, and scope of the project.

It is important to make reports easy to read. The first step is to be sure reports contain only the information needed by the recipient.

The frequency of the report submission should be decided. This varies from project to project.





Reporting Considerations

As you prepare reports, use the following guidelines:

- ❑ Be sure everything you publish is **accurate**.
- ❑ Keep all stakeholders **appropriately informed**, including team members, customers, clients, functional managers, and senior management.

Determine for each audience what information they need to perform their functions and design reports accordingly.

Keep them informed—but **not over-informed**—with the information they need to make decisions and take corrective action.

- ❑ Use **exception reporting** by including only **major variations** from the plan. Stakeholders don't have time to digest pages and pages of project information, and there is little need to report on items where status matches the plans.
- 

- 
- ❑ Choose the **best format** for the report, such as a table, line graph, histogram, or bar or Gantt chart.
 - ❑ On each report, clearly state the **purpose of the report** and the **action to be taken**.
 - ❑ Highlight main achievements
 - ❑ Report on time.
- 



Control Project Objectives

Control is the process of comparing actual performance to the baseline to determine variances, evaluate possible alternatives, and take the appropriate corrective action.

Consider the following actions to control time:

- Systematically collect performance data
 - Compare this status information with the baseline schedule.
 - Analyze variances to determine their impact.
 - Determine a course of action.
 - Take corrective action.
- 



Managing risks

Risk is anything not in the project plan that may occur and cause your project to be late, cost more or compromise its quality/performance.

Project risk is an uncertain event or condition that, if it occurs, has a negative effect on a project objective.

Risk is a potential negative impact to project.

Risks are **internal or external** events that may occur during project implementation and could threaten the achievement of project objectives and the project as a whole.





Identify risks and outline contingency measures for when they happen. This process involves three steps:

1. Identifying risks

To identify risks you can look at possible **sources of risk** or at the threats/problems that can become risks. Sources include the team members, stakeholders, sub-contractors, target groups, etc. Problems could be, for example, a change in the political environment or the loss of money through inflation.

A good way to identify relevant risks can be an **open brain-storming** session at meetings on 'What can go wrong?' Concerned ones should be involved in this process to a) raise their awareness about possible risks, and b) to identify as many relevant risks as possible





2. Assessing risks (Analyzing risks)

Once potential risks have been identified, they need to be qualified according to their **impact on the project** and their **probability of occurring**.

As with most other aspects of planning, the assessment of probability can often only be based on assumptions and educated guesses. The impact, however, can often be estimated in relation to the budget and time lost or indicators not achieved. This assessment allows projects to prioritize risks – the **'high risk' decisions** and actions have to be **taken first**.



Risk assessment matrix

	High Impact	Medium Impact	Low Impact
High Probability	Medium Risk	High Risk	High Risk
Medium Probability	Low Risk	Medium Risk	High Risk
Low Probability	Low Risk	Low Risk	Medium Risk

Project Controls

Control means to detect and rectify errors and prevent reoccurrence.

Controlling is essentially tracking and managing the core project management elements of scope, quality, time and cost.

Control keeps the whole project on-track, on time, and within budget.

Project control monitors and compares planned objectives, requirements, risks, schedules and budgets against what is actually being delivered.

Control holds people accountable and prevents small problems from becoming big ones.



Project controls provides accurate and timely project information to project management team that will enable them to make informed decisions and take necessary actions to correct any possible adverse situations or trends.

Effective project control is used to deliver timely project information allowing managers to take corrective measures to keep the project on track.

It is important to provide the right information to the right people. Data which is important for project manager could be use less for a stakeholder.





Each project should be assessed for the appropriate level of control needed. Too much control is time consuming, too little control is very risky.

There are three major points which have to be covered to build up the framework for a functioning project control system:

- What data to collect; how, when, and who will collect them?
 - Analysis of the data and
 - The report of the current status
- 

The project control Process

Basic steps of an effective and efficient project control are:

1. **Setting** a baseline plan
2. **Measuring** progress and performance
3. **Comparing** plan against actual
4. **Taking** corrective action



Human Aspects of Project Management

Of all the resources required to execute projects, **human resources is the most critical**. All other resources such as land, money, materials, machinery, and others are secondary.

Human resources are the dynamic resources, and the **success** of the project depends largely on **the availability and the quality** of the human resources and how efficiently the workforce uses other resources for project execution.





The major human resource support activities relevant to project management are

- Determining the quantity and quality of people required for project execution
 - Conducting the recruitment and training process,
 - Managing and measuring performance,
 - Compensation and rewards,
 - Managing leaves,
 - Enforcing discipline, and
 - Ensuring legal compliance.
- 



Successful project execution depends on the cooperation, commitment, and enthusiasm of the workforce to perform as required, and the ability of the project managers to manage the workforce well, to extract the best out of the available resources.



Monitoring and Evaluation (M & E)

Monitoring and evaluation occur during the execution phase of project life cycle.

Although often referred to together as M&E, monitoring and evaluation are two **different terms** but linked processes that apply to many projects.

Projects with **strong M & E** tend to **stay on track** and **problems** are often **detected earlier**, which reduces the likelihood of having major cost overruns or time delays.



Key Terms in Monitoring and Evaluation

Outputs are the **things produced** by a project or programme. For example, include products like new or rehabilitated wells and pumps, new latrines and training manuals.

Outcomes are the **effects of the outputs**, usually in the **medium-term**. Example, the number of people who now have access to safe water as a result of the new water schemes.

Impacts are **long-term effects**. Example, could be a fall in the incidence of disease or improved school attendance.

Key performance indicators (KPIs). KPIs help to measure the **time, cost, and quality** of the project.





Key Performance Indicators (KPIs)

KPIs need to be defined according to **critical or core objectives**.

KPIs cannot be arbitrarily selected by the project manager; rather, the project manager applies tools like brainstorming, interviewing, and the Delphi technique to determine what the project's stakeholders want the KPIs to be.



Examples of KPIs

- ✔ Customer satisfaction (When the project deals directly with a client or customer).
- ✔ Employee satisfaction
- ✔ Actual Cost (AC)
- ✔ Earned Value (EV)
- ✔ Cost Performance Index (CPI)
- ✔ Cost Variance (CV)
- ✔ Return on Investment (ROI)
- ✔ Planned Value (PV)- $(\text{Planned \% of tasks left to complete}) \times (\text{project budget})$
- ✔ Line Items In Budget: keep track of individual expenditures and provide a more detailed way to see how the budget was spent.
- ✔ Number of Errors: How often things need to be redone during the project.
- ✔ Number of Project Milestones Completed On Time With Sign Off: (Were the milestones completed in a timely manner and approved by the owner or buyer?)
- ✔ Cycle time (The time needed to complete a certain task or activity).



Monitoring

Monitoring is systematic, timely and purposeful **observation** and **data collection** to check if **project activities** are being **implemented as planned**.

Monitoring is **collecting, recording, and reporting** information concerning project performance.

Monitoring is **checking progress** against plans.

Monitoring is the **routine collection** and **analysis** of project data, specifically on project activities.

Data is collected **routinely** throughout the life cycle of the project. Data collection is **ongoing** (weekly, monthly, quarterly, semiannually, etc.).

It focuses on **what is being done** and **how it is being done**.





Why do we monitor?

- Simply because we know that things **don't always go** according to plan.
 - To detect and react appropriately to **deviations** and **changes** to plans.
- 

What do we monitor?

- ✓ Human resources
- ✓ Money
- ✓ Materials
- ✓ Machines
- ✓ Time
- ✓ Quality
- ✓ Tasks
- ✓ Progress, e.t.c



Monitoring provides managers with information needed to:

- Analyze the **current situation**
- Identify **problems** and find **solutions**
- Discover **trends**
- Keep project activities **on schedule**
- Measure **progress** towards objectives and formulate/revise future goals and objectives
- **Make decisions** about human, financial, and material resources.

Monitoring is usually undertaken (done) by the **people directly involved in implementing** the project. It is carried out by the beneficiaries, the implementing staff, the supervisory staff and the management staff.





Characteristics of an excellent monitoring system

- Quickly provides information for corrective action
 - Cost-effective
 - Flexible
 - Accurate
 - Comprehensive
 - Relevant
 - Accessible
 - Leads to learning
 - Transparent
 - Shares information up and down.
- 



Monitoring reports

The purpose of a project monitoring report is to provide information to assist stakeholders in **comparing** performance against **plans** so that potential problems can be identified and analyzed.

Reports are only **effective** if they are submitted to the **right people** at the **right time** to facilitate **corrective decision making**.





A monitoring report should:

- Contain a **list of the activities** to be monitored (derived from the plan),
- List the **duration** and **deadlines** for completion of different activities,
- State the **methods of monitoring** the activities,
- State the **current progress** on steps taken so far,
- Point out any **changes** in the original goals, objectives or activities and explain this change in direction.
- State the **barriers** confronted, if any,
- Suggest **solutions** to overcome them.

The data acquired through monitoring is **used for evaluation**.



Evaluation

Evaluation is defined as an objective and rigorous analysis of a continuing or completed project, to determine its **significance, effectiveness, impact** and **sustainability** by comparing the result with the set of standards.

The evaluation of a project is usually carried out **periodically** (mid-term, final or post- evaluation).

Evaluations should help to **draw conclusions** about five main aspects of the intervention:

- Relevance
- Effectiveness
- Efficiency
- Impact
- Sustainability

Evaluation looks for **lessons to be learned** from both success and lack of success, and also looks for **best practices** which can be applied elsewhere.

Reflections on lessons learned from past successes and failures can be very helpful when planning future interventions.

Types of Evaluation

Types of evaluation include:

- **Interim evaluation** (Mid-Term Evaluation) -This normally takes place at some point **during the life** of a project, usually **mid-term**.
- **Terminal evaluation** - This assesses the progress made towards the achievement of the pre-determined objectives **at the end** of the project and provides a **basis for decisions** on future action. Its findings and recommendations are often used to **decide whether or not to stop** the project or when a new phase is under consideration.
- **Ex-post evaluation** - This is conducted **after** a sufficient **number of years** (depending on the project) have elapsed since project completion so as to **measure the impact**. It is generally accepted that, if the evaluations are to be objective, they have to be undertaken by **external consultants**.



Types of Evaluation

- **Internal Evaluation**

Evaluation conducted by a unit and/or individuals reporting to the management of the donor, partner, or implementing organization.

- **Joint Evaluation**

An evaluation to which different partners and/or donor agencies participate.

- **Participatory Evaluation**

Evaluation method in which **representatives** of agencies and stakeholders (including **beneficiaries**) work together in designing, carrying out and interpreting an evaluation.





Types of Evaluation

- **Formative Evaluation**

Evaluation intended to **improve performance**, most often conducted during the implementation phase.

- **Summative Evaluation**

Evaluation conducted **at the end** to determine the extent to which anticipated **outcomes** were produced.

Summative evaluation is intended to provide information about the **worth** of the project.





Evaluation can be **internally** or **externally** led, but is **not usually facilitated** by those who **implement** the project. Evaluation is best conducted by an independent outsider who can be impartial in consulting with project staff.



Key Differences Between Monitoring and Evaluation

The difference between monitoring and evaluation can be drawn clearly on the following premises:

- ❑ By monitoring is meant a **routine process** (continuous activity), that scrutinizes the **activities and progress** of the project and also **finds out the deviations** that occur while undertaking the project. As against, evaluation is a **periodical activity** that makes inferences (draw conclusion) about the **relevance and effectiveness** of the project or program.
- ❑ While monitoring is **observational** in nature, evaluation is **judgmental**.

Monitoring is conducted by **internal party** whereas evaluation is conducted by **internal party or external party** (who can give impartial views on the project).

- ❑ Monitoring is a **short-term process**, that is concerned with the collection of information regarding the success of the project. Conversely, evaluation is a **long-term process**, which not only records the information but also assesses the outcomes and impact of the project.



Link between planning, M&E

Without proper planning, it is **not clear** what should be monitored and how.

Without effective planning, the basis for evaluation is **weak**.

Without careful monitoring, the necessary data is not collected; hence evaluation **cannot be done well**.

Monitoring is necessary for evaluation. Monitoring facilitates evaluation.



The Prerequisites for successful project implementation are as follows:

➤ Adequate formulation.

- Good assessment of input requirements
- Deep field investigation
- Using sound procedures for computing costs and benefits
- Omission of project linkages
- Accurate estimation of benefits and estimating costs

➤ **Sound project organization.**

- Led by competent leader
 - Authority and responsibility are equitably distributed
 - Adequate attention is paid to designing jobs
 - All work methods and systems are clearly defined
 - A culture of reward and punishment on the basis of performance is established
- 

➤ **Proper implementation planning.**

- Estimate the resource requirements and time plan for each activity
- Define inter linkage between different activities of the project
- Specify cost standards

➤ **Advance action.**

- Acquisition of land
 - Security essentials clearance
 - Identify technology collaborators/consultants
 - Arranging infrastructure facilities
 - Preliminary design and engineering
 - Calling of tenders
- 



➤ **Timely availability of funds.**

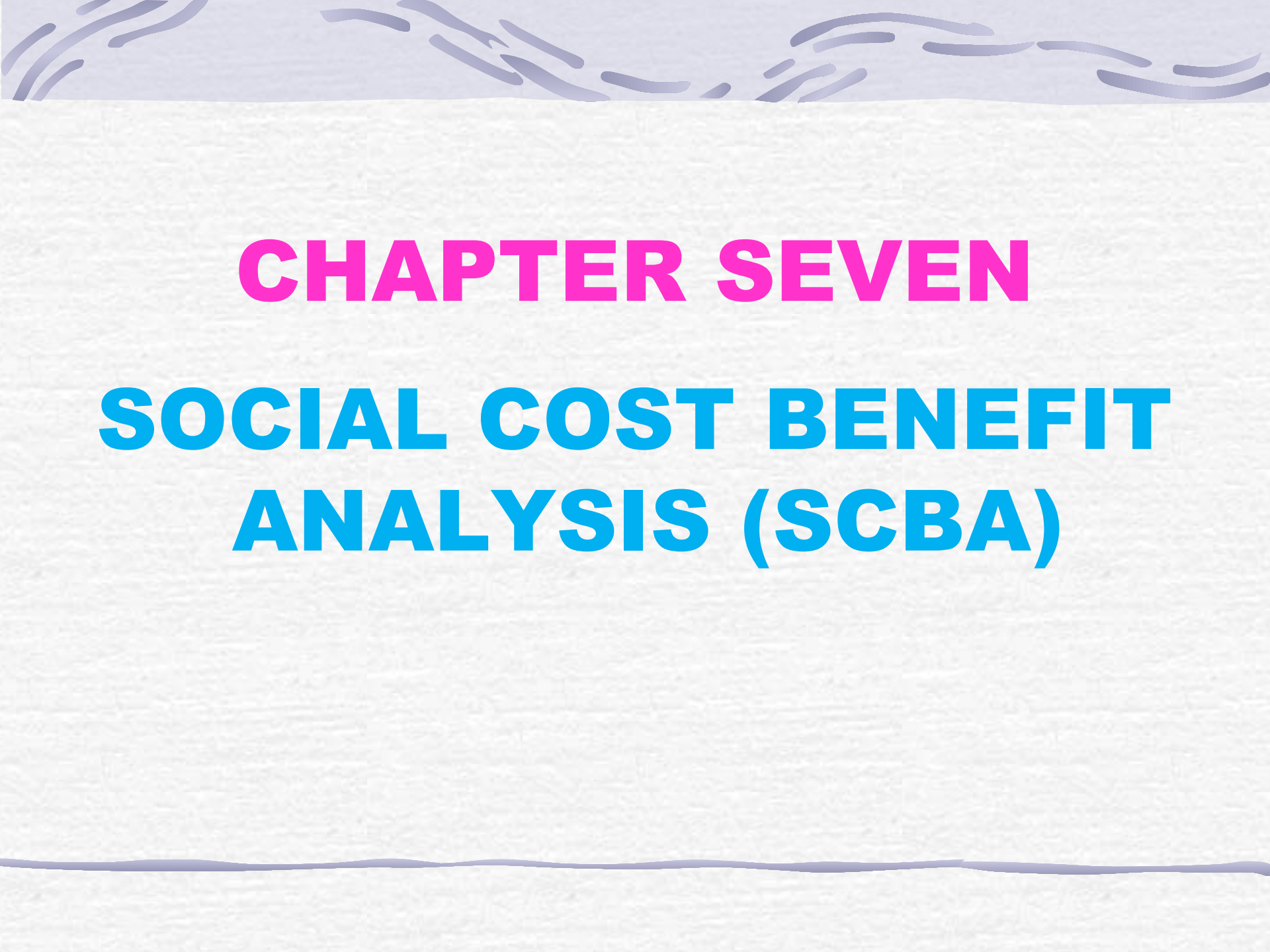
- In order to be successfully completed the project must have adequate funds available to meet its requirements as per the implementation plan. Proper allocation of must be done with the help of experts.

➤ **Better contract management.**

- Competence and capability of contractors must be ensured before entering into a contract.
- Project authorities must retain the power to transfer a contract to third parties when delays are anticipated.

➤ **Effective monitoring.**

- Adequate system of monitoring must be established.
- 



CHAPTER SEVEN

SOCIAL COST BENEFIT ANALYSIS (SCBA)



Social cost benefit analysis

SCBA is also called **Economic analysis**. It is a part of the economic analysis in project appraisal.

SCBA is a methodology developed for **evaluating investment projects** from the point of view of the **society (or economy)** as a whole.

All project investments have impact on society and nations economy.





In this analysis, the direct economic **benefits** and **cost** of the project on

- **Distribution of income** in society,
 - Level of **savings and investments** in society,
 - The contribution of the project towards fulfillment of certain **merit-wants** (e.g. employment, self-sufficiency etc.) are analyzed.
- 



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- 



Social costs

Social cost of staff

- Lay off and voluntary termination
- Work-related injuries
- Undesired attitude formation

Social cost of community and general public

- Increase in cost of living in vicinity due to a project
 - Consumption of state electric service
 - Services and facilities consumed
 - Expenditure in foreign exchange
 - Environmental damage
- 



Examples of Social cost

- Air pollution
 - Water pollution
 - Soil erosion
 - Depletion and destruction of animal resource
 - Deforestation
 - Impairment of human resources
- 



Social benefits of staff

- Medical and hospital facilities
 - Educational facilities
 - Subsidized canteen facilities
 - Provident fund
 - Housing facilities
 - Holiday, leaves
 - Recreational and cultural facilities
- 



Social benefits of community and general public

- Local taxes paid to municipality
 - Environmental improvements
 - Generation of job potential
 - Welfare activity for the community
 - Business generation
- 



Example

If a bridge is to be constructed then **how much will it benefit the people** who live in that particular area, is to be analyzed.

Therefore, how many people are **willing to use** the bridge, how much **traffic will be reduced** and what is the **increase in cost** of travelling, will have to be assessed as a whole to come to a conclusion.

Suppose, if people are **not willing to use** the bridge if the cost of travelling from the bridge is \$5 and if \$7 has to be charged per vehicle to make this project feasible, then the government may consider **dropping the project out**.





On the other hand, if people are willing to travel using the bridge, being indifferent to the toll price-difference of \$2, and the traffic is reduced by a good amount, then the government will sanction the project.

Therefore, it is beneficial to take up a project if its total benefits (B) are more than its total costs. Thus, it can be said that if, $B/C > 1$ or even when $B=C$, then a project can be under taken.





SCBA can be applied to **public** investment and also to **private** investment.

In case of **public investment**, it plays a major role in the **economic development** of a developing country. SCBA is **used** primarily for **evaluating public investments** especially in developing countries where government play significant role in economic development.

In case of **private investments** social cost benefit analysis is important as investments are to be **approved** and are **monitored** by the government.





Advantages of SCBA

- The ability to identify the projects that maximize the welfare of the country.
 - The ability to objectively assess and quantify the purpose projects in relation to community needs.
 - Ability to rank and prioritize limited resources so that the maximum benefit is realized.
- 



Disadvantages

- **Difficulty in measuring** social costs and benefits and converting them in to monetary term.
 - **Over statement** of the value of social benefits
 - **Complexity**
 - **Conflict** between social welfare and financial justification.
- 



CHAPTER EIGHT

PROJECT FINANCING



Every project **needs financing** to implement and run it successfully.

Project finance is nothing but **sourcing funds** to a long term infrastructure project, or any other project, and **using the cash flow** generated from the project to **payback** the financing procured.

Project finance is the **long-term financing** of infrastructure and industrial projects based upon the projected **cash flows of the project** rather than the balance sheets of its sponsors.

The **debt and equity** used to finance the project are paid back from the cash flow **generated by the project**.





In project finance, financial institutions can't see your balance sheet in case of a project. They finance the project on the basis of the projected cash flow. If the cash flow seems satisfactory and beneficial to the financial institutions they invest in the project.





The most visible characteristic of project finance is that it is **non-recourse** debt.

Non-recourse loans are **secured** by the **project assets** and **paid entirely** from **project cash flow**, rather than from the general assets or creditworthiness of the project sponsors.

In non-recourse financing, the borrowers and shareholders of the borrower **have no personal liability** in the event of monetary default.

If the collateral sells for less than the debt, the lender **cannot seek that deficiency balance** from the borrower—its recovery is limited only to the value of the collateral.

 Project finance is especially attractive to the private sector because companies can fund major projects off-balance-sheet.

Parties involved:

There are many parties involved in project finance. Let's look at all these parties in brief

✔ **Sponsors:** People who sponsor the project.

✔ **Lender:** Financial institutions which lend money for project.

✔ **Financial advisors:** They help the parties to understand how much return on investment they can make. They can be in both sides – lenders or borrowers.

✔ **Technical advisors:** Often for effective execution of the project, technical consultants are hired. They act as technical advisors for the project.

✔ **Legal advisors:** As the name suggests, they help in legal matters.

✔ **Debt Financier:** People who give secured loan for the project on the basis of project assets.

✔ **Equity Investors:** People who invest the money in lieu of shares.

✔ **Regulatory agencies:** Generally, government authorities who take care of the regulations in regard to the project intricacies.

✔ **Multilateral agencies:** The agencies are part of the World Bank group.



Sources of Project Financing

There are a wide variety of funding sources available for projects.

The main sources of project finance include

- ✓ **Equity**
 - ✓ **Debt**
 - ✓ **Grants**
 - ✓ **Lease and hire-purchase**
- 

✓ Equity

Equity refers to **capital invested** by sponsor(s) of the project and others.

Equity is provided by project sponsors, government, third party private investors, and internally generated cash.

Equity providers require a **rate of return** target, which is **higher than** the interest rate of debt financing. This is to compensate the **higher risks** taken by equity investors as they have junior claim to income and assets of the project.



✓ Debt

Debt refers to **borrowed capital** from banks and other financial institutions. It has **fixed maturity** and a **fixed rate** of interest is paid on the principal.

Lenders of debt capital have **prior claim** on income and assets of the project.

The **common forms of debt** are:

- Commercial loan
 - Bridge finance
 - Bonds
- 



Commercial loans are funds lent by **commercial banks** and **other financial institutions** and are usually the main source of debt financing.

Bridge financing is a **short-term financing** arrangement (e.g., for the construction period or for an initial period) which is generally **used until a long-term** financing arrangement can be implemented.

Bonds are long-term interest bearing debt instruments.



✓ **Government grants**

Government grants can be made available to make projects commercially viable, to **reduce the financial risks of private investors**, and to achieve socially desirable objectives such as to **induce economic growth** in lagging or disadvantaged areas.

✓ **Lease and hire-purchase**

Leasing for industrial equipments and machineries is a relatively recent phenomenon.

A **lease** represents a contractual arrangement whereby the lessor grants the lessee the **right to use an asset** in return for periodical lease **rental payments**.

Hire purchase involves the **purchase of an asset** on the understanding that the purchaser (called hirer) will **pay in equal periodic installments** spread over a length of time.